TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

id0110q2801000

Operation After Replacing Timing Chain

1. After replacing the timing chain, perform the following procedure.

STEP	ACTION	PAGE/CONDITION
1	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
2	Switch the ignition off.	—
3	Verify that the check engine light does not illuminate.	—
4	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
5	Perform fuel injector injection amount correction.	(See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].)
6	Perform timing chain learning procedure.	(See TIMING CHAIN LEARNING [SKYACTIV-D 1.5].)
7	Clear the DTCs.	(See OBD READ/CLEAR DIAGNOSTIC TEST RESULTS [SKYACTIV-D 1.5].)
8	Switch the ignition off.	—

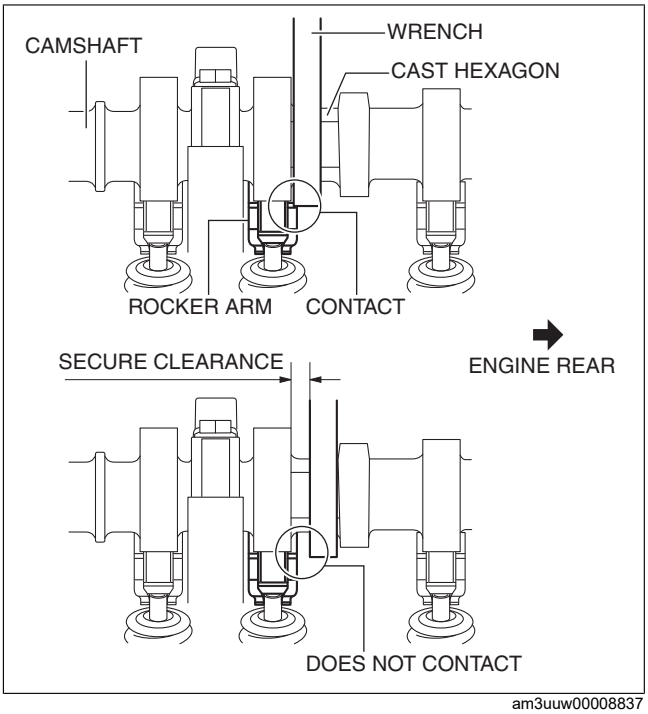
Timing Chain Removal/Installation

Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

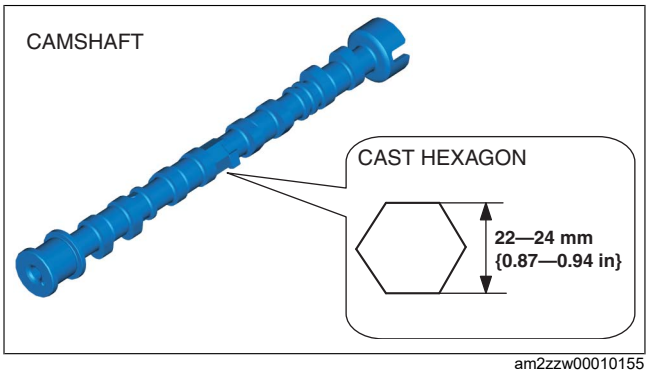
Caution

- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**



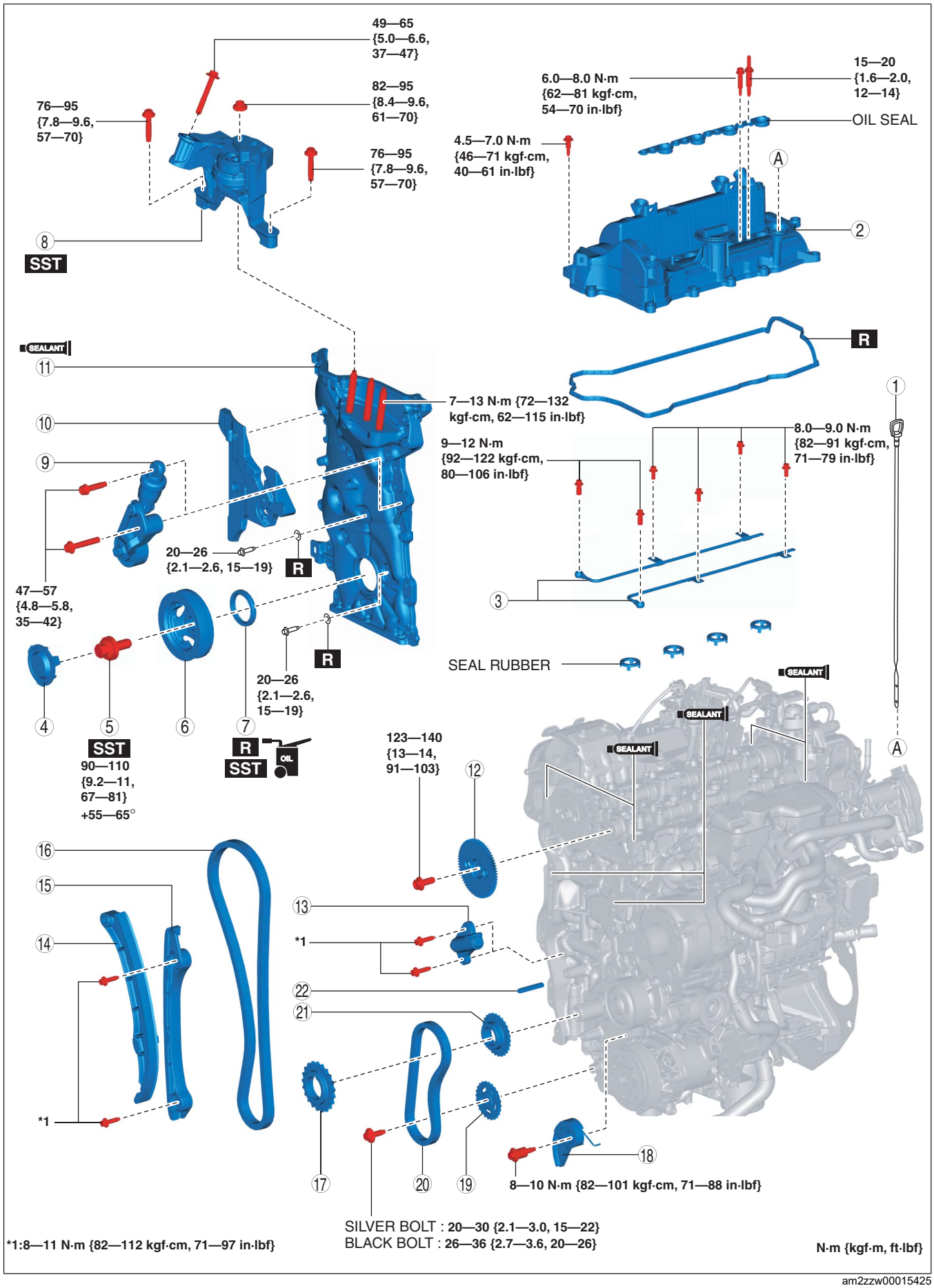
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Note
- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}.**



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1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Remove the front splash shield. (RH) (See SPLASH SHIELD REMOVAL/INSTALLATION.)
5. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Drain the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].)
7. Remove the oil pan. (See OIL PAN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8. Remove the fuel injectors. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9. Remove the PCM and PCM bracket as a single unit. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
10. Remove in the order indicated in the table.
11. Install in the reverse order of removal.
12. Refill with the specified type and amount of the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].)
13. If the timing chain is replaced, perform “Operation After Replacing Timing Chain”. (See Operation After Replacing Timing Chain.)
14. Start the engine and inspect the following:
    - Leakage of engine oil.
    - Oil level (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].)
    - Runout and contact of pulley and belt.
    - Idle speed (See ENGINE TUNE-UP [SKYACTIV-D 1.5].)



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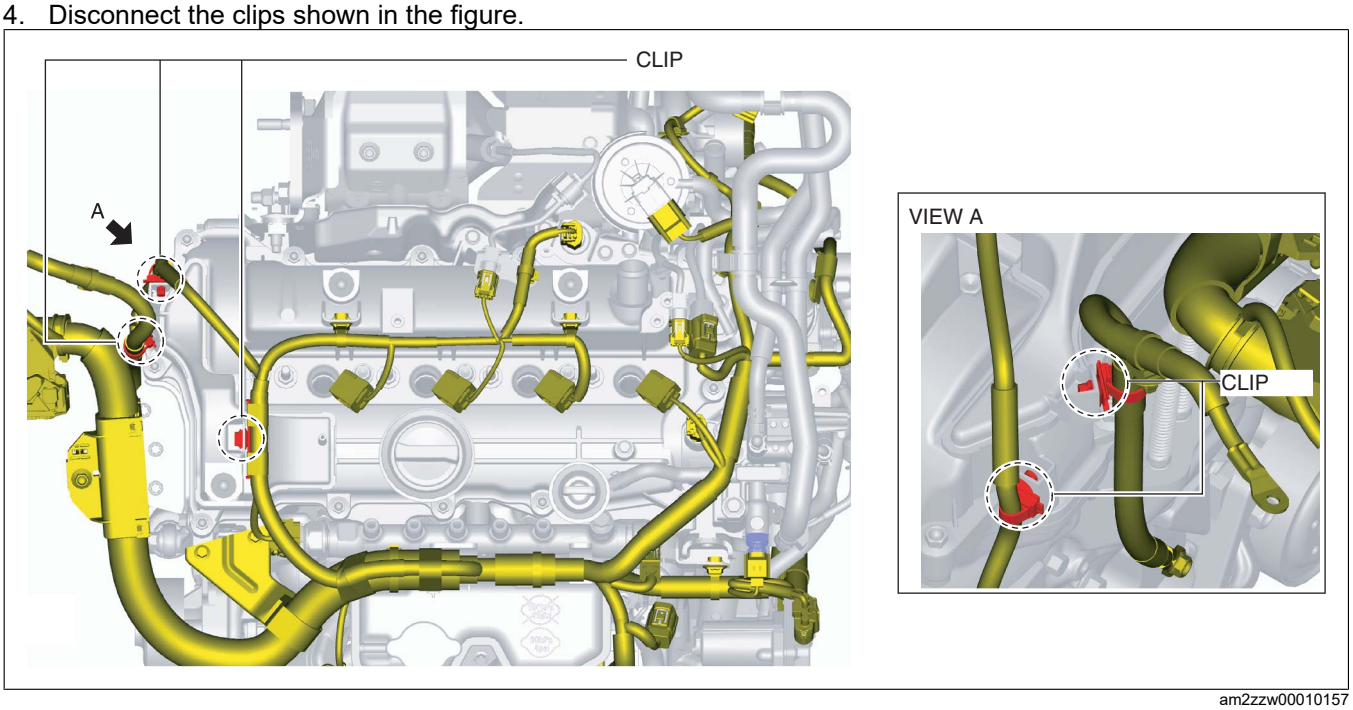
1	Dipstick

2	Cylinder head cover (See Cylinder head cover removal note.) (See Cylinder head cover installation note.)
3	Oil shower pipe (See Oil shower pipe installation note.)
4	Crankshaft pulley cover (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].)
5	Crankshaft pulley lock bolt (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].)
6	Crankshaft pulley (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].)
7	Front oil seal (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].)
8	No.3 engine mount (See No.3 engine mount removal note.) (See No.3 engine mount installation note.)
9	Drive belt auto tensioner (See DRIVE BELT AUTO TENSIONER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
10	Seal rubber (See Seal rubber removal note.) (See Seal rubber installation note.)
11	Engine front cover (See Engine front cover removal note.) (See Engine front cover installation note.)

12	Intake camshaft sprocket (See Intake camshaft sprocket removal note.) (See Intake camshaft sprocket installation note.)
13	Timing chain tensioner (See Timing chain removal note.) (See Timing chain installation note.)
14	Timing chain tensioner arm (See Timing chain removal note.) (See Timing chain installation note.)
15	Timing chain guide (See Timing chain removal note.) (See Timing chain installation note.)
16	Timing chain (See Timing chain removal note.) (See Timing chain installation note.)
17	Crankshaft sprocket (See Timing chain removal note.) (See Timing chain installation note.)
18	Oil pump chain tensioner (See Oil pump chain tensioner installation note.)
19	Oil pump driven sprocket (See Oil pump driven sprocket removal note.) (See Oil pump driven sprocket installation note.)
20	Oil pump chain
21	Oil pump drive sprocket
22	Key

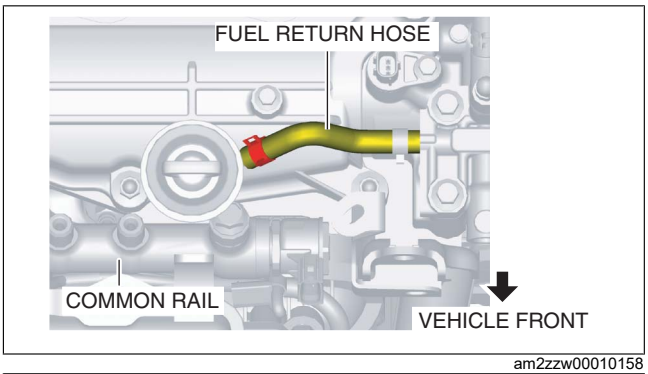
Cylinder head cover removal note

1. Remove the injection pipe (supply pump side). (See INJECTION PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Disconnect the breather hose from the cylinder head cover. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the vacuum pipe. (See SUPPLY PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Disconnect the clips shown in the figure.



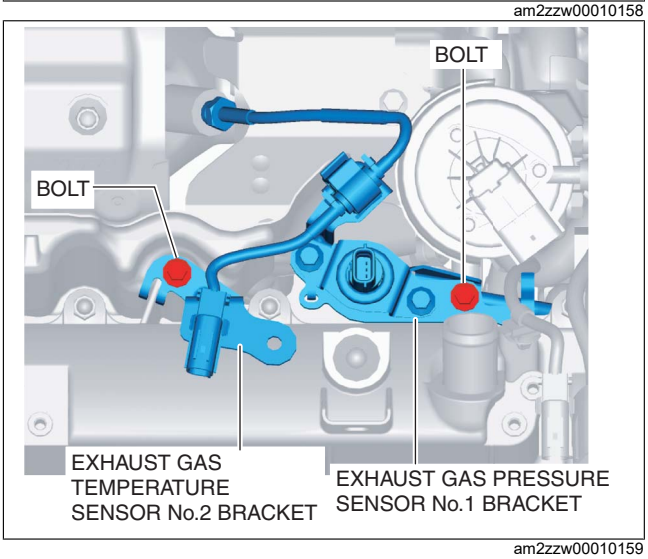
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5. Disconnect the fuel return hose from the common rail. (See COMMON RAIL REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)



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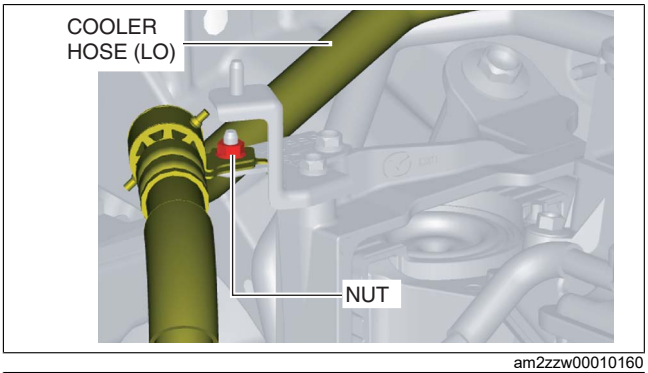
6. Remove the bolts for the exhaust gas pressure sensor brackets No.1 and exhaust gas temperature sensor No.2 and set them aside at the rear of the vehicle.
7. Remove the cylinder head cover.



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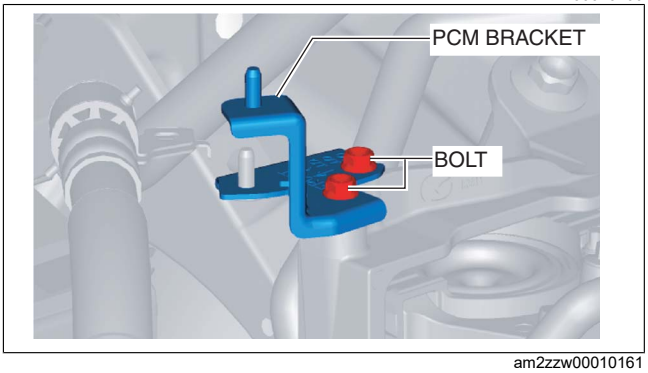
No.3 engine mount removal note

1. Remove the nut shown in the figure and set the cooler hose (LO) aside.



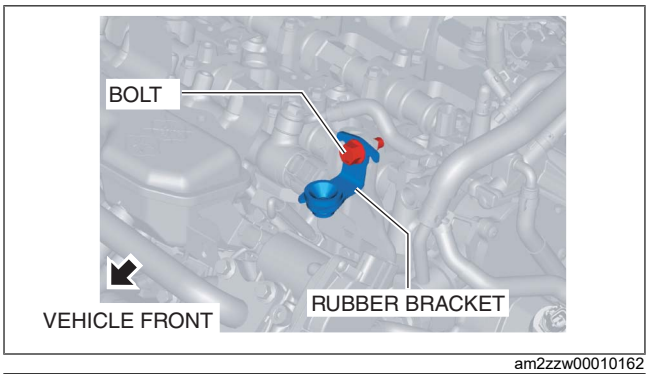
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2. Remove the bolts shown in the figure and remove the PCM bracket.



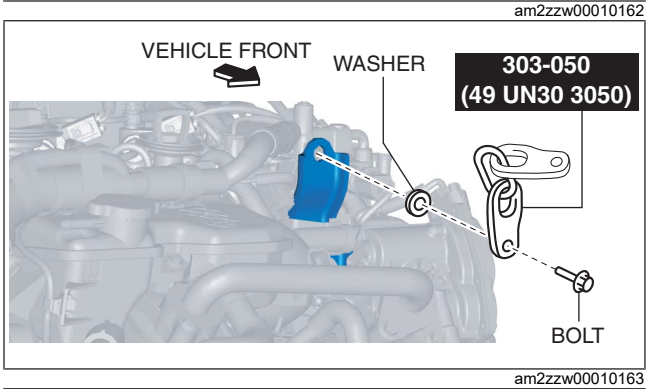
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3. Remove the rubber bracket.



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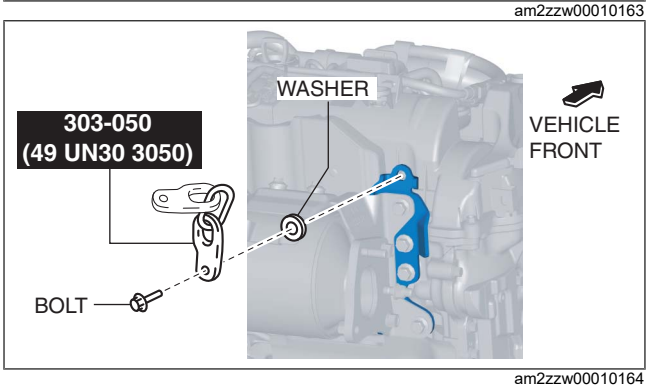
4. Install the **SST** to the position shown in the figure using the following bolt and washer.



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Bolt:
part number 99794 1025 or an M10 × 1.25, length 25 mm {0.98 in}
Washer:
approx. 3 mm {0.1 in} thickness

Tightening torque
38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}

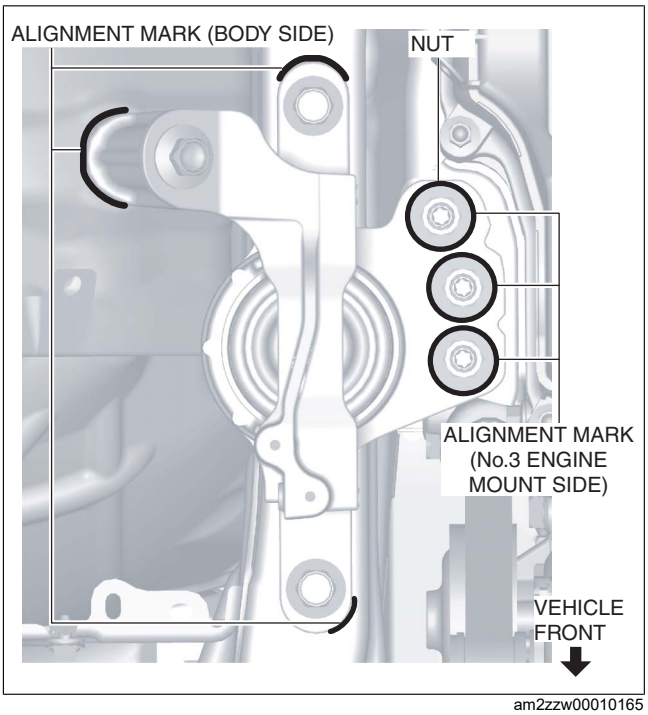


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Caution
- **Slots have been adopted for the No.3 engine mount installation holes. If the No.3 engine mount is deviated from the original position when installing the No.3 engine mount, engine noise or vibration could increase. Before removing the No. 3 engine mount, place alignment marks on the No.3 engine mount and body so that they can be assembled to the same positions as before removal.**

5. Place alignment marks on the locations shown in the figure so that they can be assembled to the same positions as before removal.

Note
- Paint so that the No.3 engine mount is framed on the body side and the outline of the nut is framed on the No.3 engine mount side.

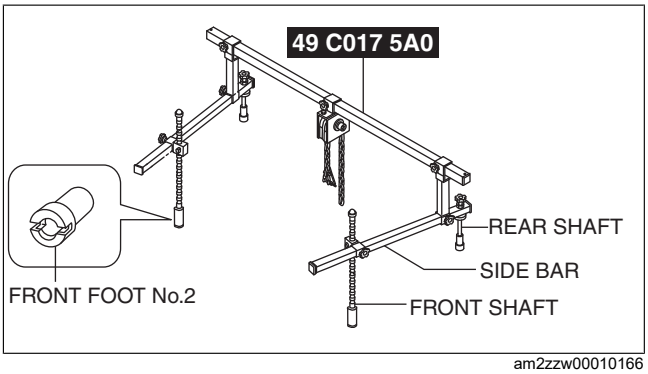


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6. Install the **SST** using the following procedures.

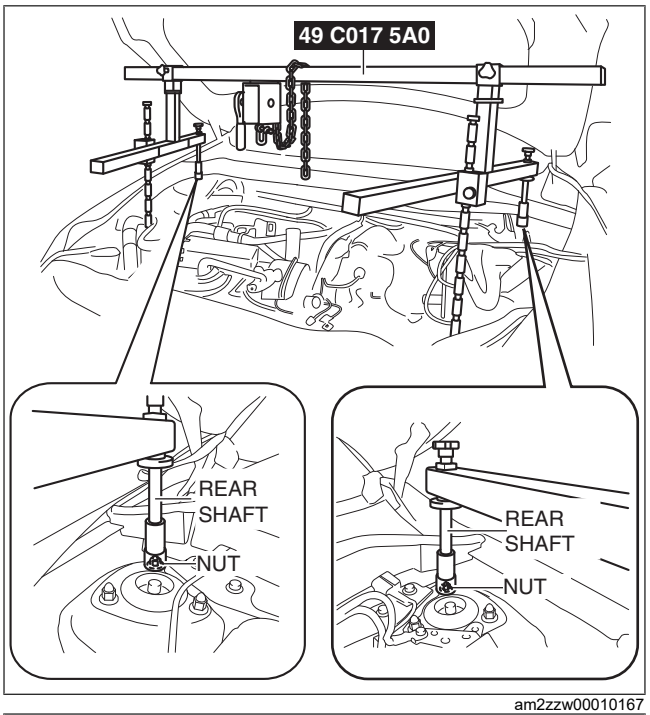
Caution
- **Refer to the SST instruction manual for the basic handing procedure.**

Note
- Install front foot No. 2 to the left and right front shafts of the **SST**.



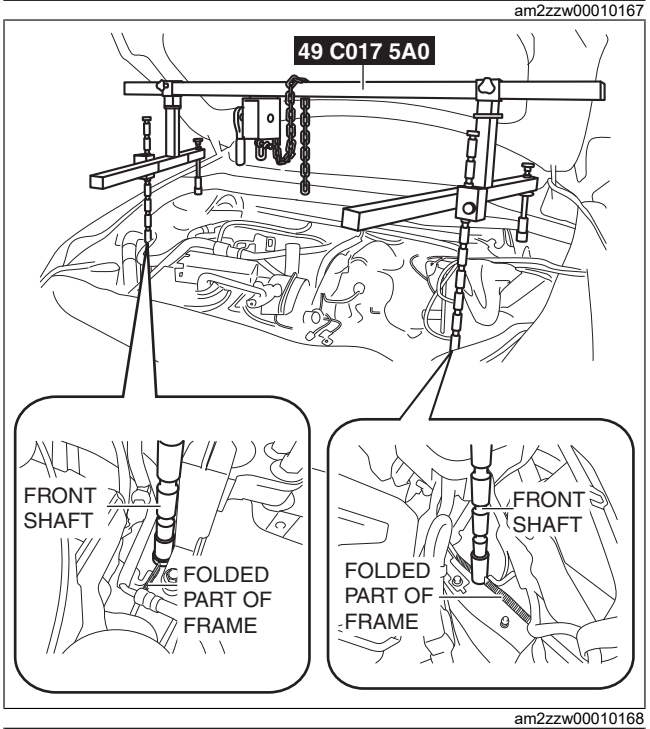
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(1) Set the rear shafts of the **SST (49 C017 5A0)** on the left/right front shock absorber nuts (top) as shown in the figure.



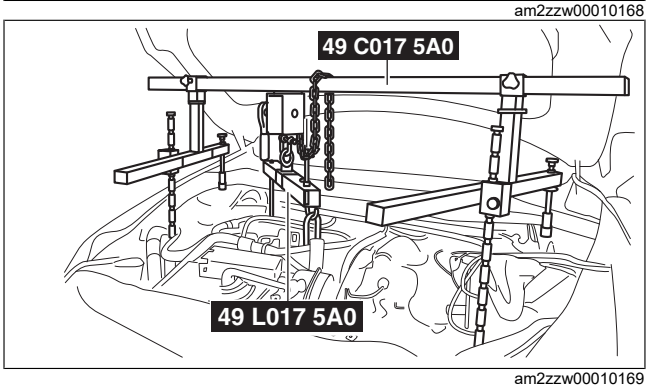
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(2) Set the front shafts of the **SST (49 C017 5A0)** on the bent up part of the frame as shown in the figure.



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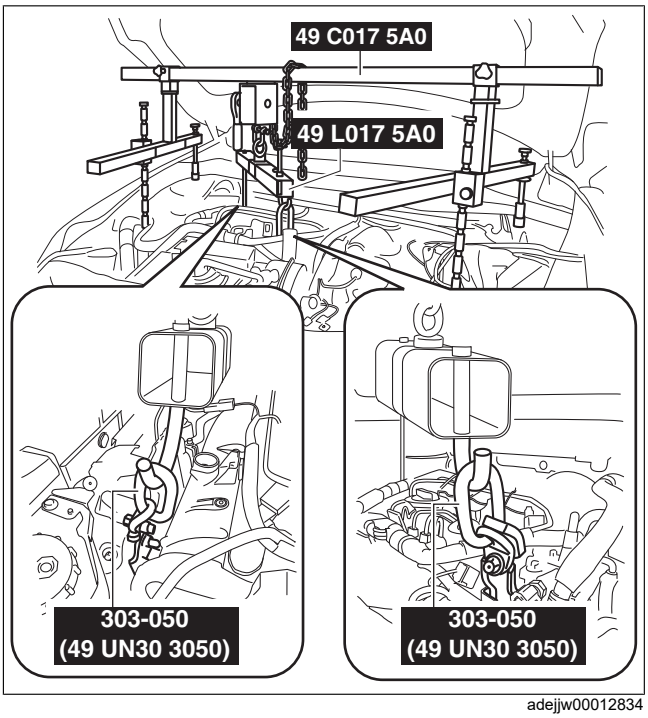
(3) Install the **SST (49 L017 5A0)** to the **SST (49 C017 5A0)** as shown in the figure.



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(4) Install the **SST (49 L017 5A0)** to the **SST (49 UN30 3050)** with the hook of the **SST (49 L017 5A0)** facing outward.
(5) Adjust the left and right sides of the **SST (49 C017 5A0)** so that the height of the side bars is even, and tighten each part of the **SST (49 C017 5A0)**.
(6) Apply tension to the chain to support the engine and verify that the engine is securely hung.

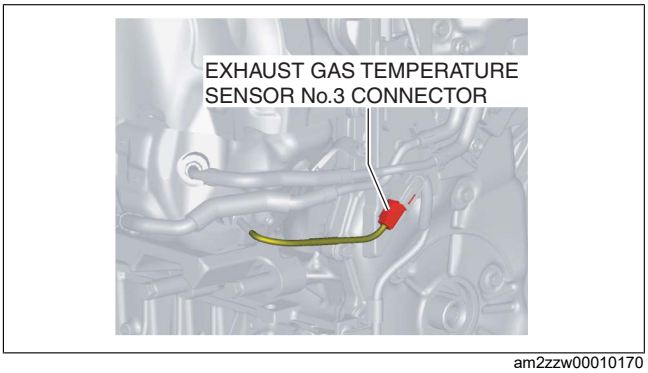
7. Remove the No.3 engine mount.



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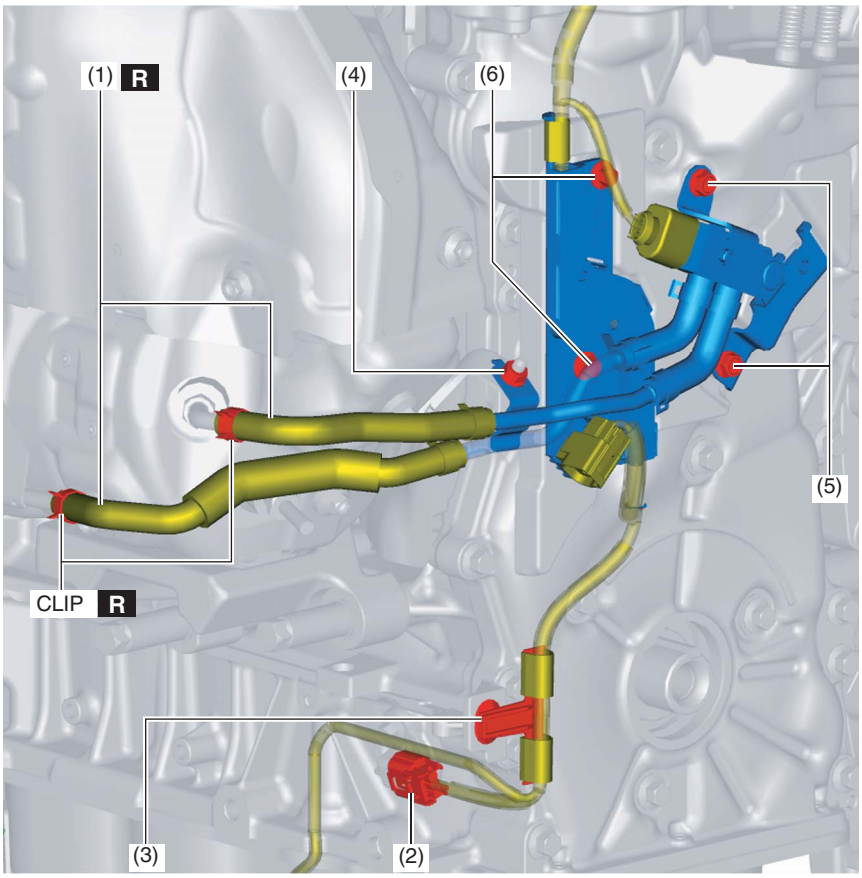
Seal rubber removal note

1. Disconnect the exhaust temperature sensor No.3 connector.



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2. Disconnect or remove the following parts in the order of the numbers shown in the figure.



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(1) Air hoses
(2) Crankshaft position sensor connector
(3) Wiring harness clip
(4) Air pipe stay installation nut
(5) Exhaust gas pressure sensor No.2 bracket installation bolts
(6) Wiring harness protector installation bolts

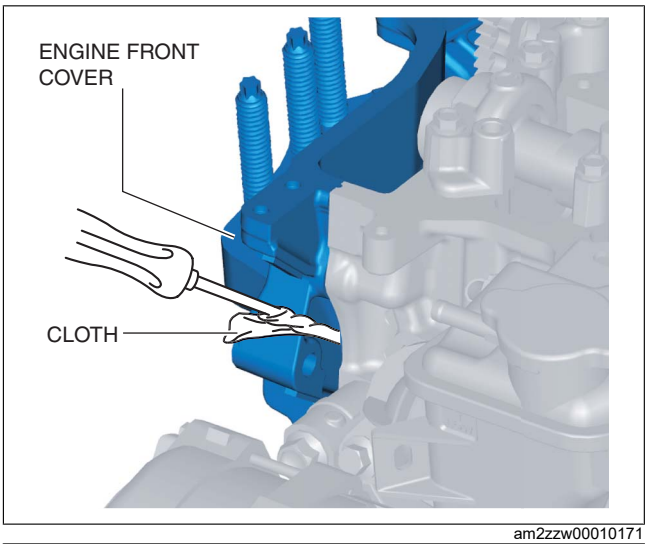
3. Set the wiring harness, air hose component, and exhaust gas pressure sensor No.2 on the engine as a single unit.
4. Remove the seal rubber.

Engine front cover removal note

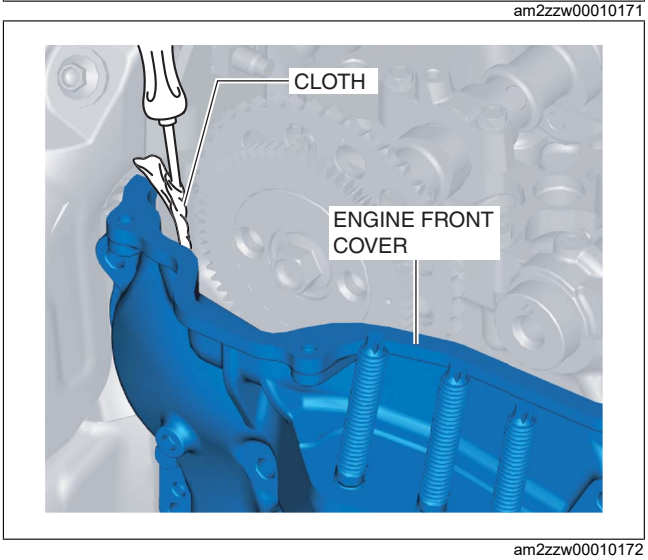
1. Remove the engine front cover installation bolt.
2. Using a screwdriver wrapped in a cloth, peel the sealant away a little at a time, and remove the engine front cover.

Caution

- **Do not apply excessive force to the screwdriver. Otherwise, the engine front cover could be damaged.**
- **Be careful not to scratch or damage the seal surface. Otherwise, it could cause oil leakage.**



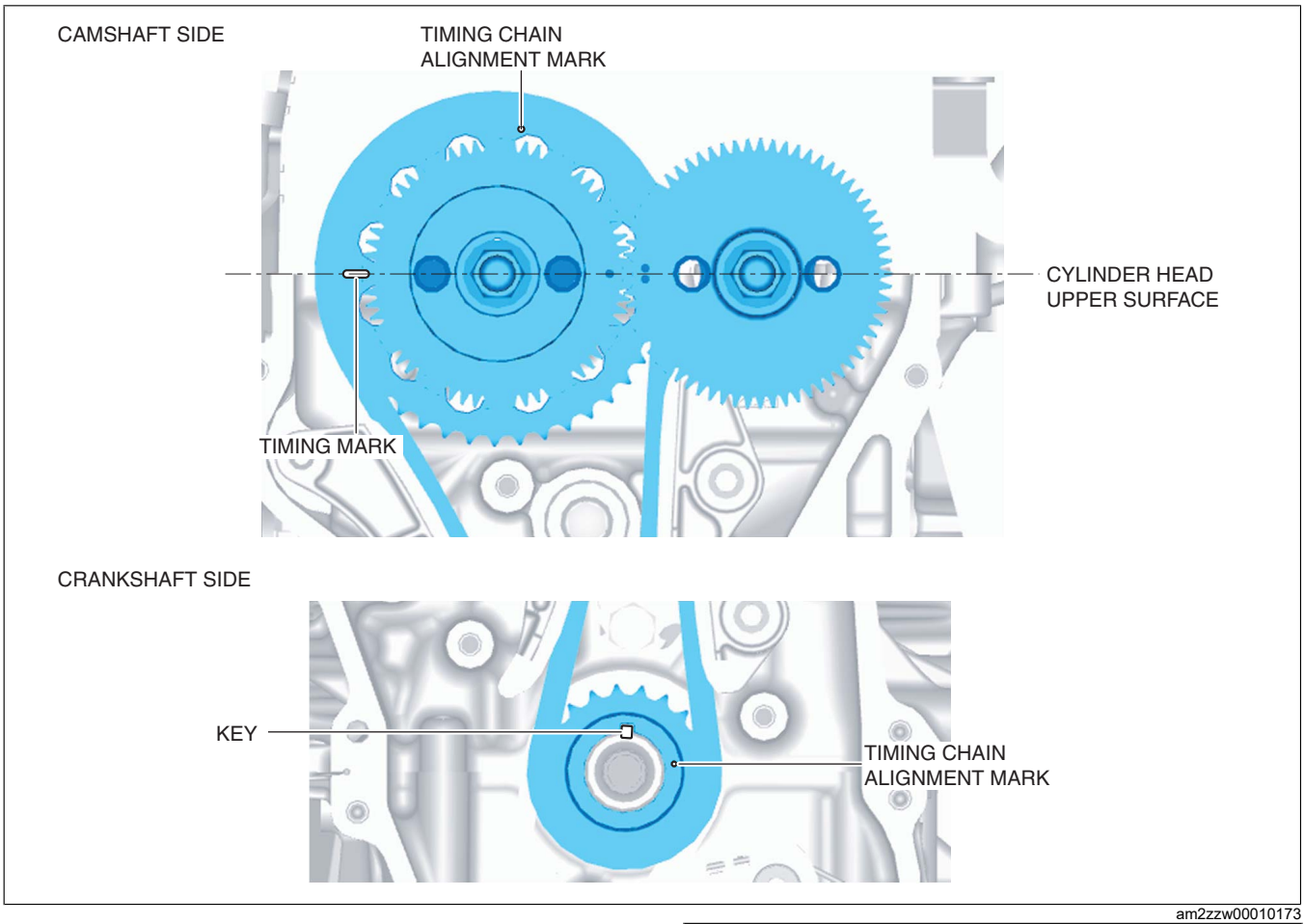
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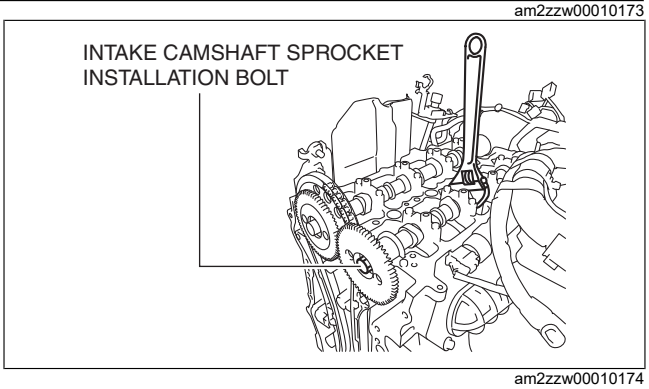
Intake camshaft sprocket removal note

1. Rotate the crankshaft clockwise to align the timing mark and the key position as shown in the figure, and set cylinder No.1 at top dead center (TDC).



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2. Hold the intake camshaft using a wrench on the cast hexagon, and remove the intake camshaft sprocket installation bolt.
3. Remove the intake camshaft sprocket.

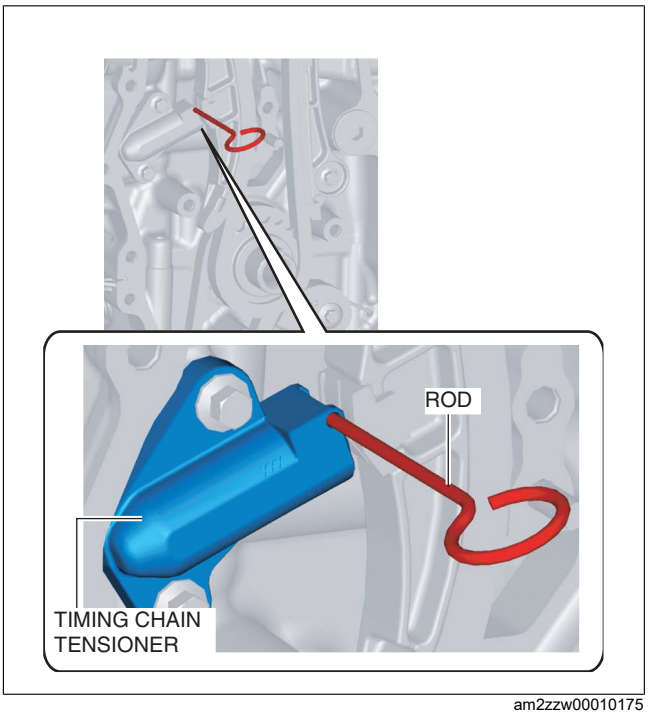


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Timing chain removal note

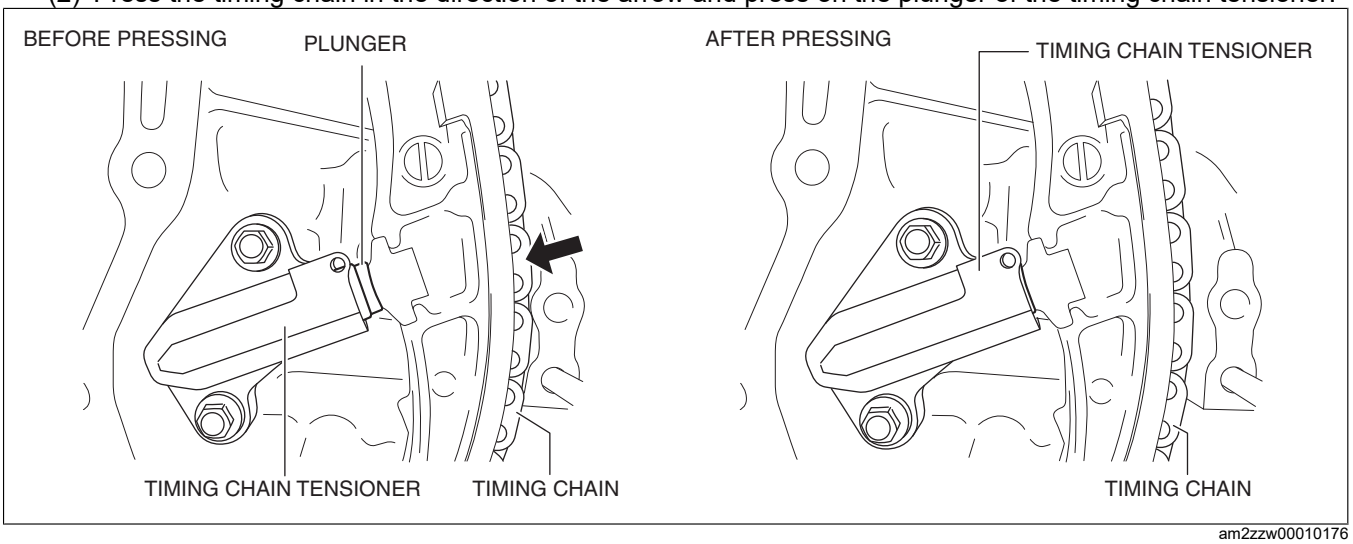
1. Loosen the timing chain tensioner using the following procedure:

(1) Insert a metal rod **(diameter approx. 3.2 mm {0.13 in}, length approx. 60—70 mm {2.4—2.7 in})** into the hole in the body of the timing chain tensioner.



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(2) Press the timing chain in the direction of the arrow and press on the plunger of the timing chain tensioner.



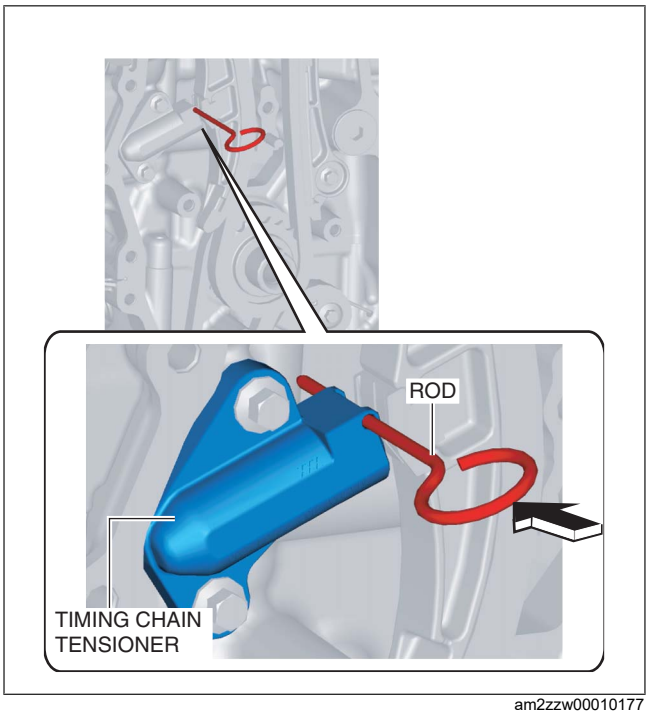
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(3) With the plunger pressed in, further insert the metal rod set in (1) above.

Note
- The rod secures the plunger and releases the tension.

(4) Loosen the power of the hand pressing the plunger and verify that the pressed-in rod does not move.

2. Remove the timing chain tensioner and timing chain tensioner arm.
3. Remove the timing chain guide.
4. Remove the timing chain and crankshaft sprocket as a single unit.

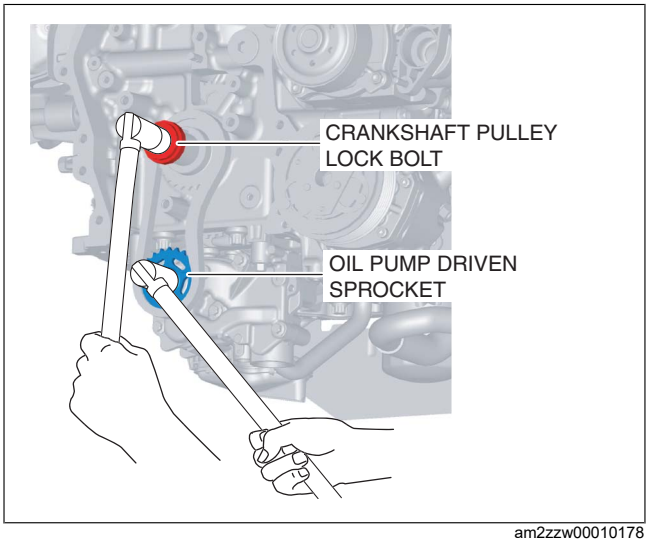


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Oil pump driven sprocket removal note

Caution
- **If the oil pump driven sprocket is removed with the oil pump chain tensioner installed, the tensioner could be damaged. Always verify that the oil pump chain tensioner is removed.**

1. Temporarily assemble the crankshaft pulley lock bolt, and lock the oil pump against rotation as shown in the figure.
2. Remove the oil pump driven sprocket.
3. Remove the temporarily assembled crankshaft pulley lock bolt.



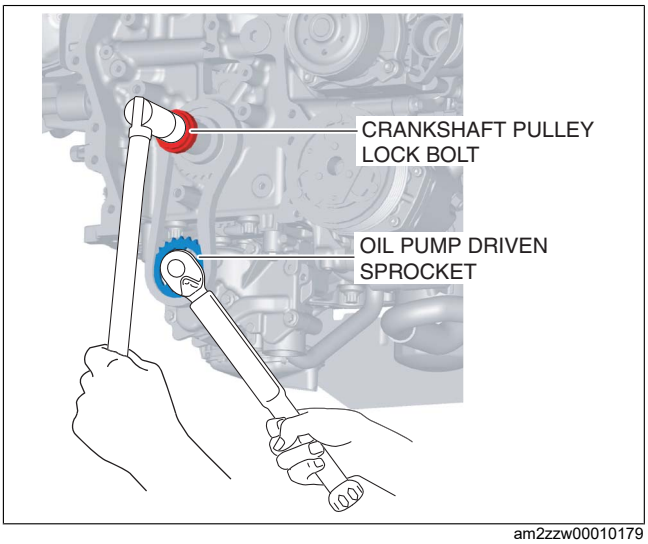
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Oil pump driven sprocket installation note

1. Temporarily assemble the crankshaft pulley lock bolt, and lock the oil pump against rotation as shown in the figure.
2. Install the oil pump driven sprocket.

   Tightening torque
   Silver bolt: 20—30 N·m {2.1—3.0 kgf·m, 15—22 ft·lbf}
   Black bolt: 26—36 N·m {2.7—3.6 kgf·m, 20—26 ft·lbf}

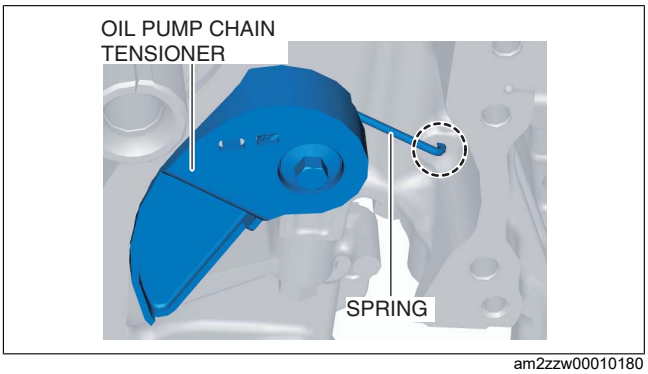
3. Remove the temporarily assembled crankshaft pulley lock bolt.



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Oil pump chain tensioner installation note

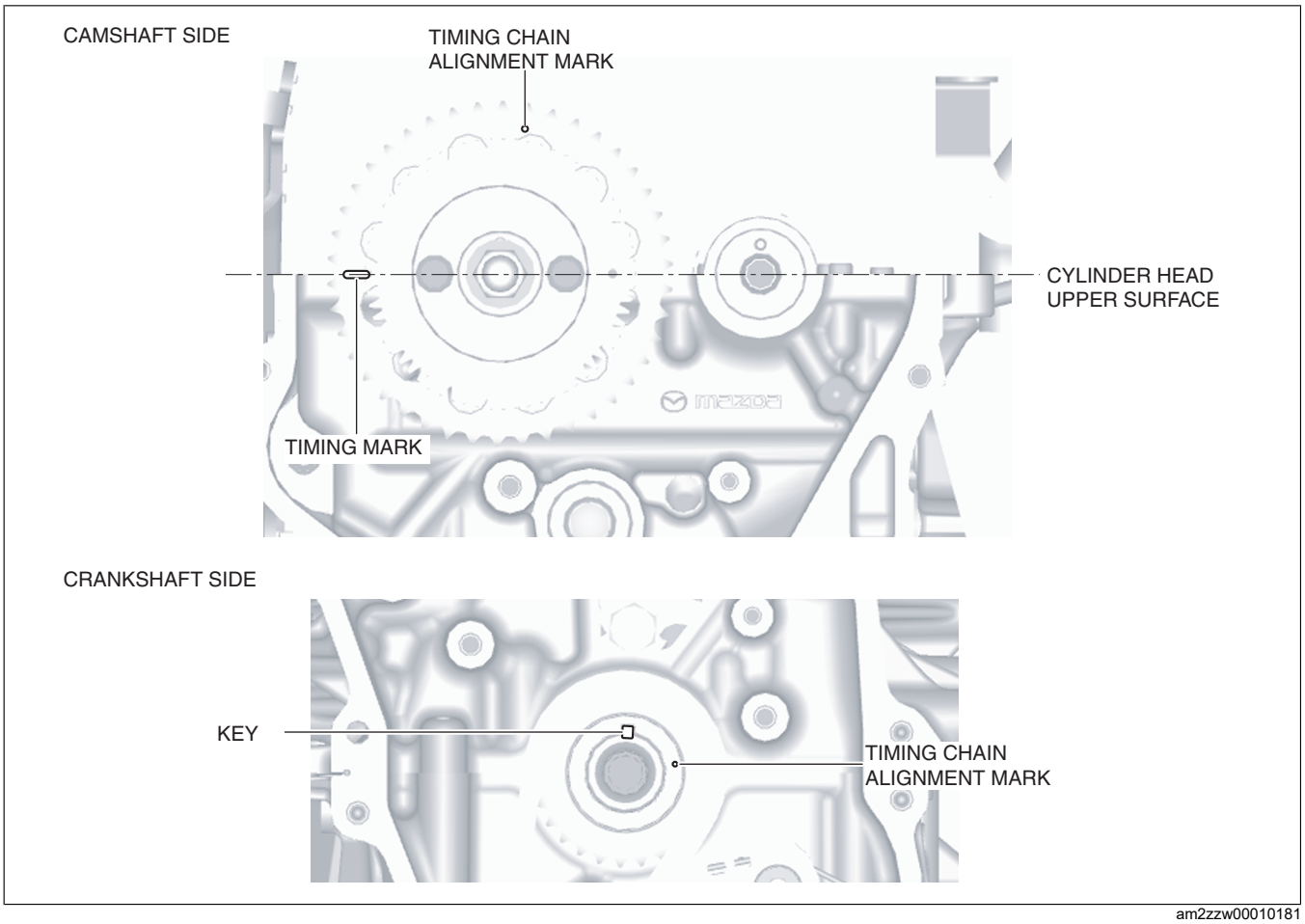
1. Temporarily assemble the oil pump chain tensioner.
2. Verify that the spring of the oil pump chain tensioner is in the hole of the lower cylinder block.
   - If the spring is not in the hole, perform the temporary assembly of the oil pump chain tensioner again.
3. Tighten the oil pump chain tensioner installation bolt.



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Timing chain installation note

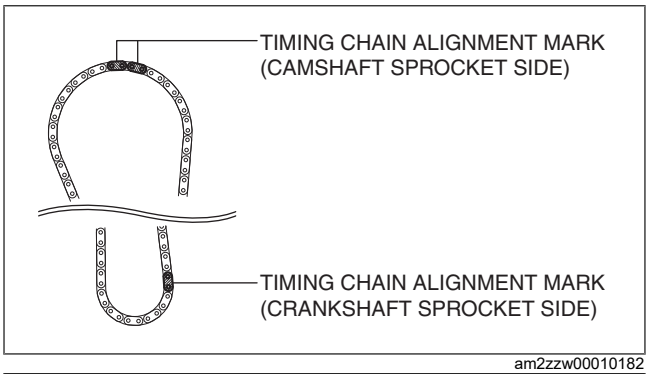
1. Verify that the timing marks and the key are aligned to the position shown in the figure.



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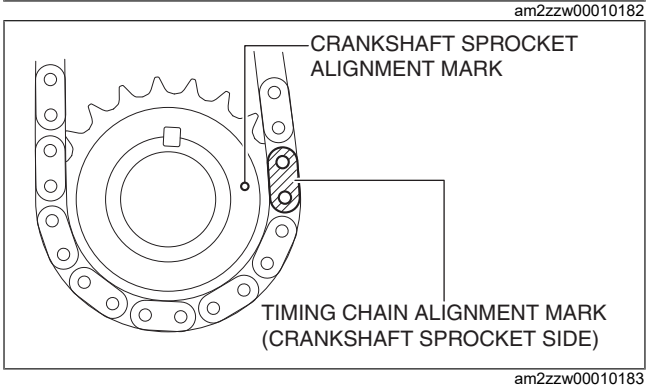
Note

- If they are not in the positions shown in the figure, rotate the camshaft and crankshaft to set the cylinder No.1 top dead center (TDC).
- There are two timing chain alignment marks as shown in the figure.



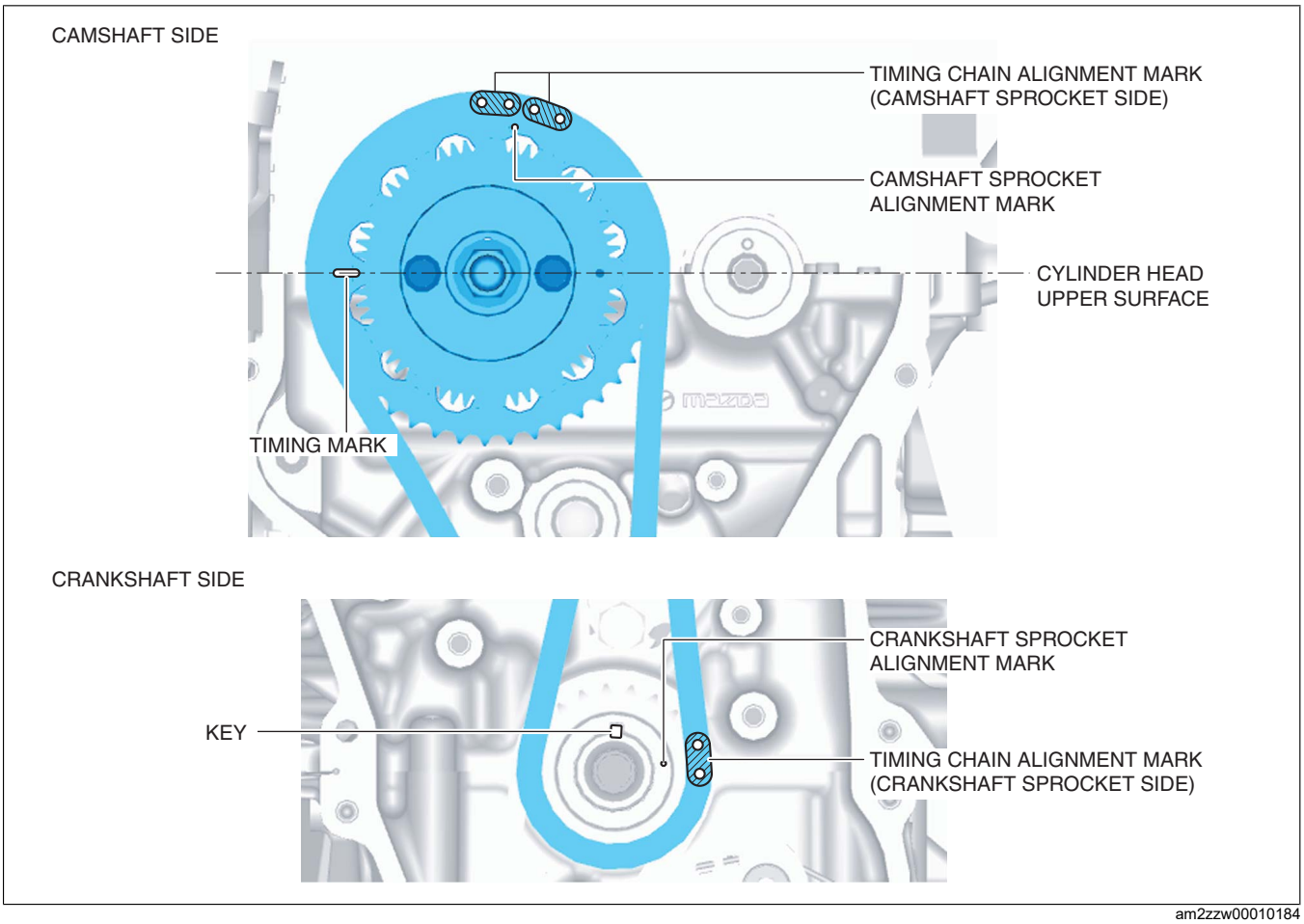
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2. Align the timing chain alignment mark (crankshaft sprocket side) with the crankshaft sprocket alignment mark.



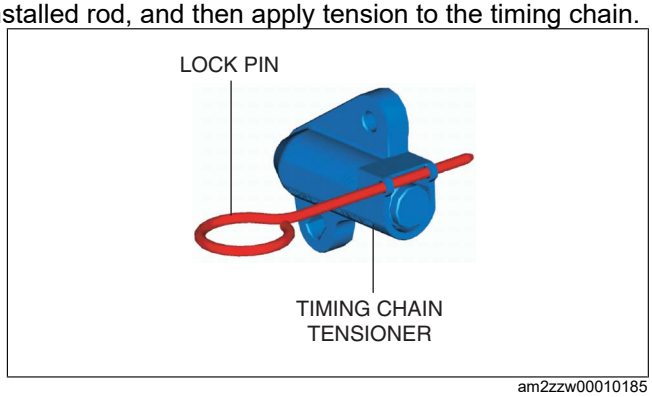
am2zzw00010183

3. Install the timing chain and crankshaft sprocket as a single unit while aligning the marks on each sprocket and the timing chain as shown in the figure.



am2zzw00010184

4. Install the timing chain guide.
5. Install the timing chain tensioner arm.
6. Install the timing chain tensioner.
7. After installing the timing chain tensioner, remove the installed rod, and then apply tension to the timing chain.
   - If a new timing chain tensioner is used, remove the installed lock pin.
8. Verify that there is no looseness in the timing chain, and re-verify that each sprocket is in the specified location.

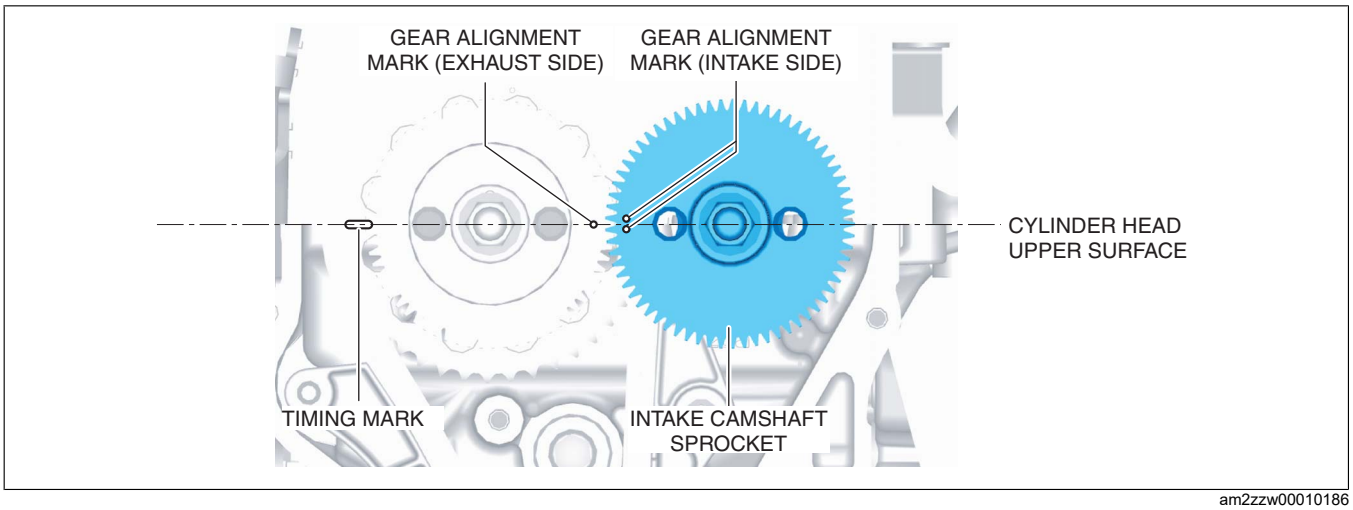


am2zzw00010185

Intake camshaft sprocket installation note

Caution

- **Align the gear alignment marks and assemble the sprocket.**

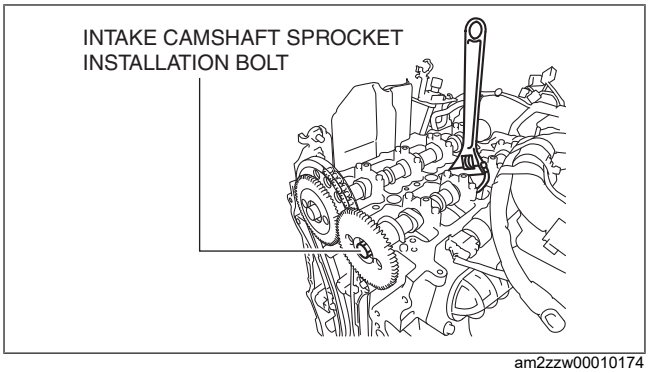


am2zzw00010186

1. Align the knock pin on the end of the intake camshaft with the knock pin installation hole on the intake camshaft sprocket side, and temporarily assemble the bolt.

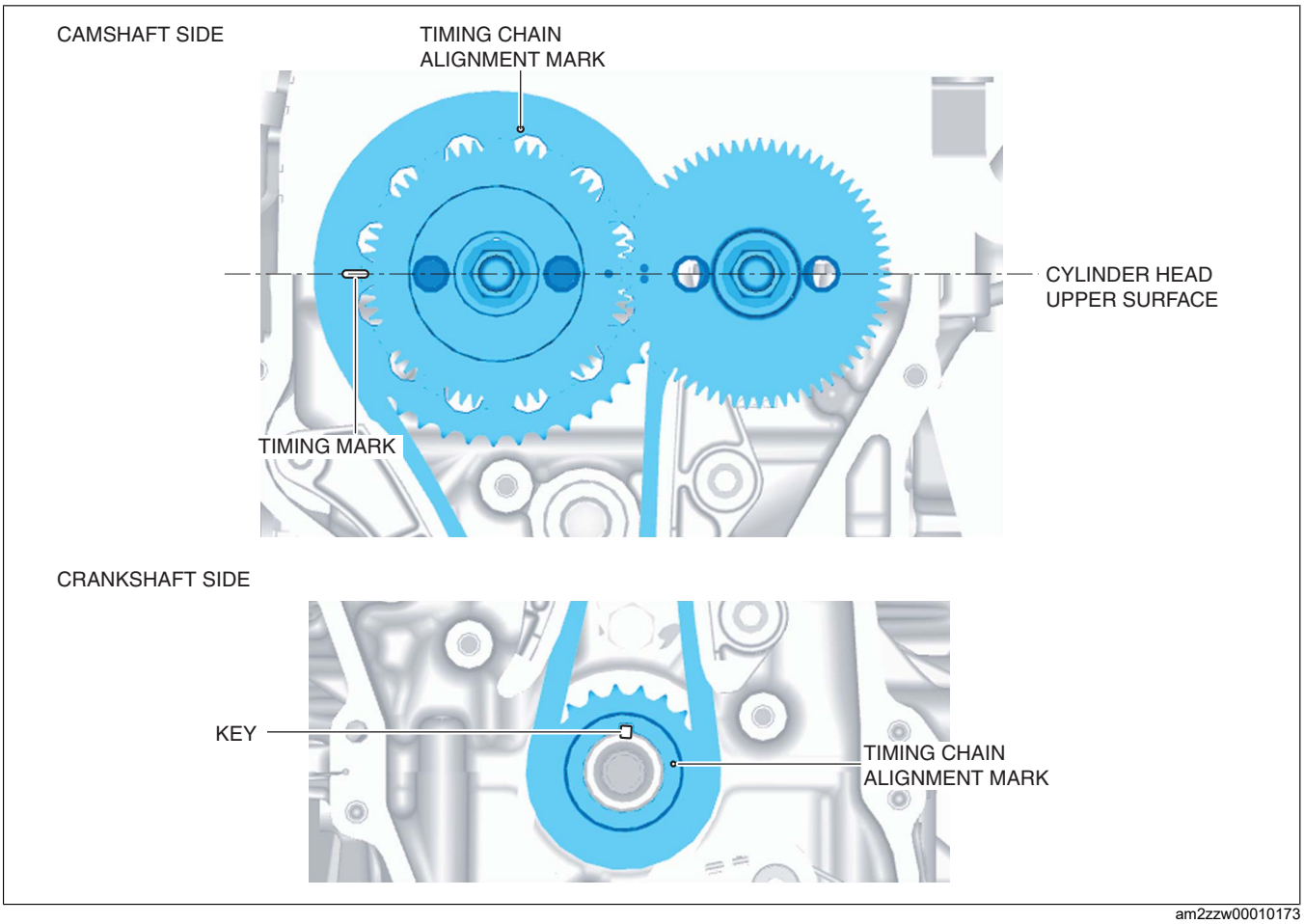
   Note
   - Finely adjust the knock pin position by rotating the camshaft and align the knock pin with the installation hole.

2. Hold the intake camshaft using a wrench on the cast hexagon, and tighten the intake camshaft sprocket installation bolt.



am2zzw00010174

3. Rotate the crankshaft clockwise **two turns** and inspect the valve timing.

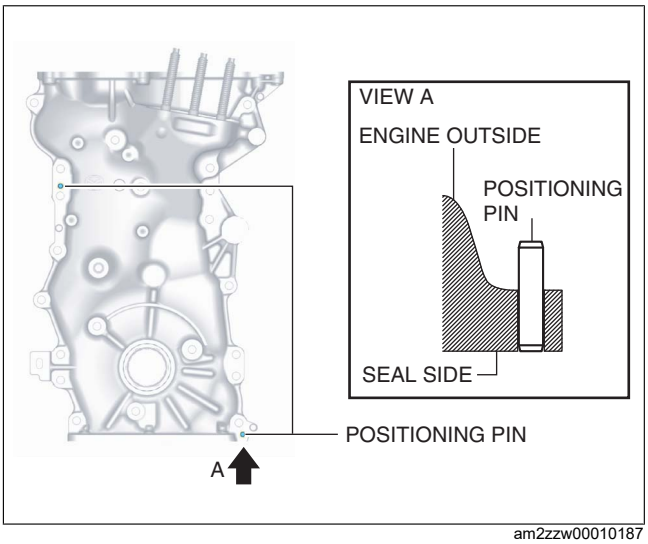


am2zzw00010173

Engine front cover installation note

Note

- For a new engine front cover, the positioning pins in the two locations shown in the figure project to the outside of the engine.



am2zzw00010187

1. If the engine front cover is newly replaced, tap the positioning pins in the two locations to the seal surface side.

   Caution
   - **If a bolt with sealant adhering to it is used, it could result in cracks in the cylinder head and cylinder block.**

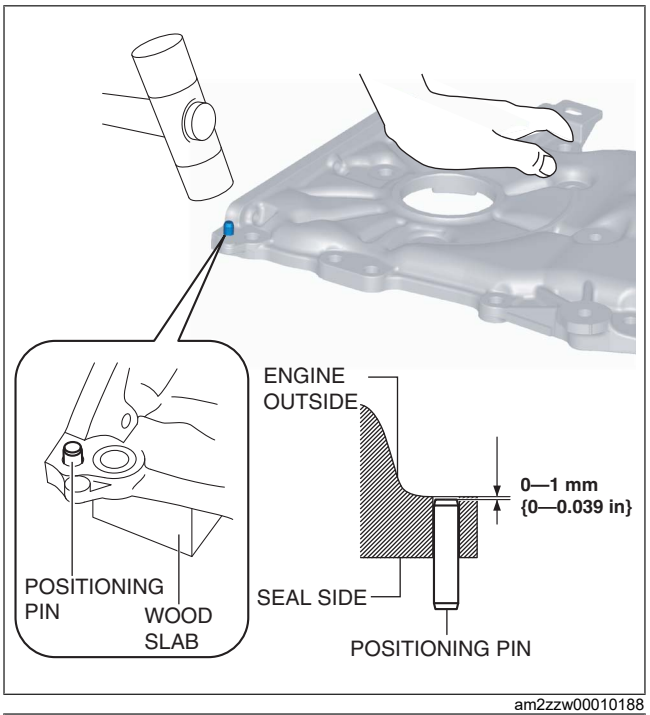
2. When reusing an engine front cover installation bolts, remove sealant adhering to the bolts.

   Caution
   - **If oil, dirt and sealant remains on the sealant application area, the silicone sealant will not seal which will cause oil leakage.**

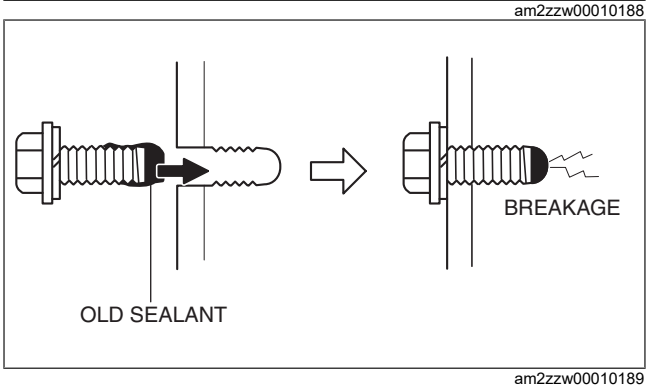
3. Completely clean and remove any oil, dirt, sealant or other foreign matter that may be adhering to the engine front cover, cylinder head, and cylinder block.

   Caution
   - **Apply the silicon sealant in a single, unbroken line.**
   - **To prevent silicone sealant from hardening, adhere the engine front cover and the cylinder block firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

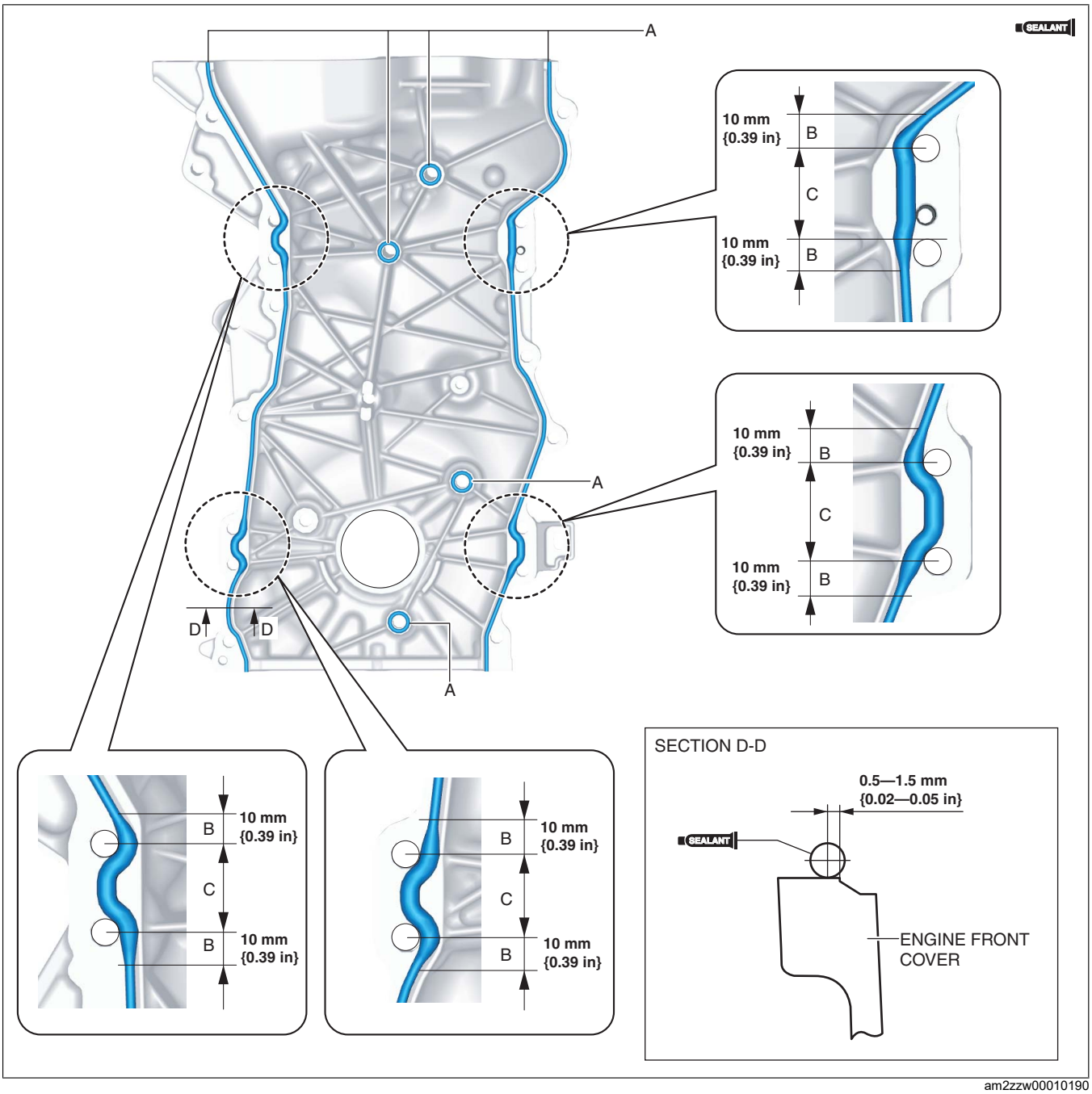
4. Apply silicone sealant (TB1217D or equivalent) to the engine front cover as shown in the figure.



am2zzw00010188



am2zzw00010189



am2zzw00010190

Bead thickness

A: 2—6 mm {0.1—0.2 in}
B: 4—6 mm {0.16—0.23 in}
C: 4—8 mm {0.2—0.3 in}

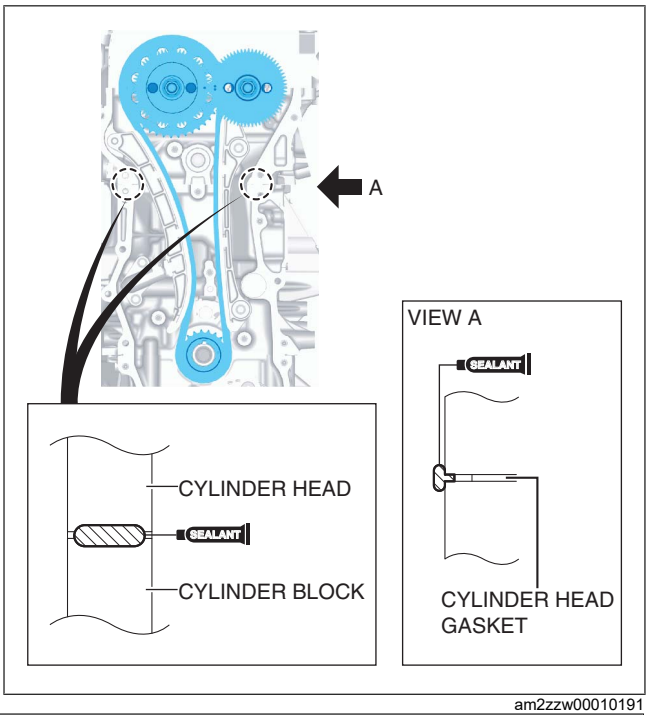
5. Apply silicone sealant to the areas shown in the figure.

Caution
- **Apply the silicone sealant so that it goes into the cylinder head gasket.**

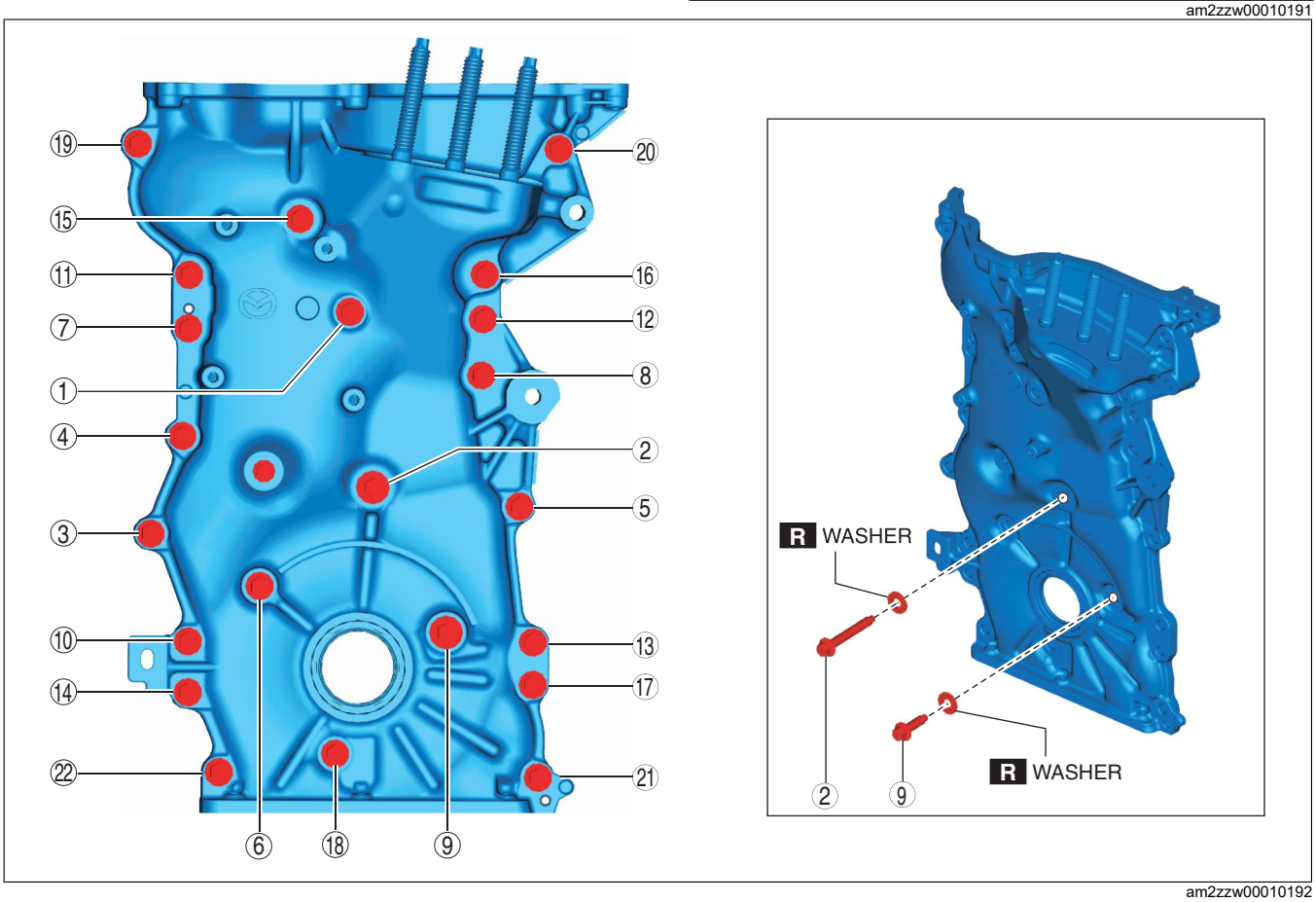
6. Install the engine front cover to the engine.

Caution
- **For the number 2 and 9 bolts of the tightening order, install the bolts with new washers.**

7. Tighten the engine front cover installation bolts in the order shown in the figure.



am2zzw00010191



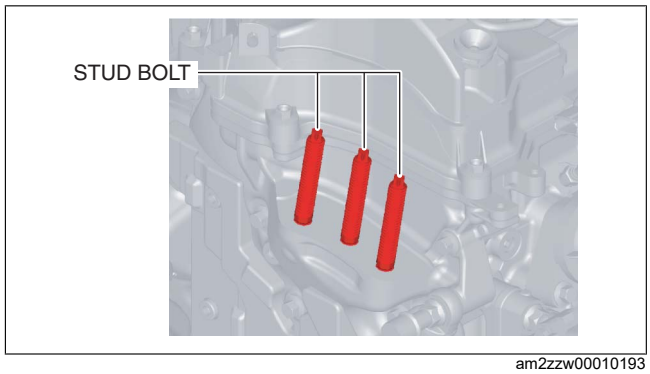
am2zzw00010192

Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}

Note
- Install stud bolts because there are no stud bolts installed on a new engine front cover.
- If the engine front cover is reused, tighten the stud bolts because they may have loosened.

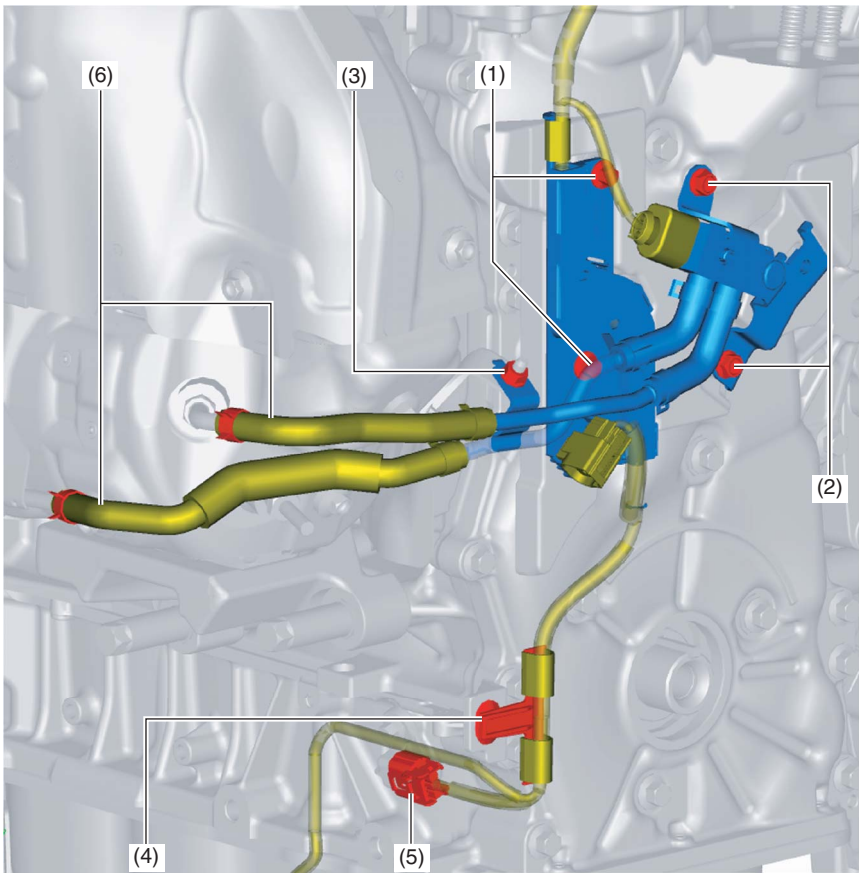
8. Tighten the engine front cover stud bolts.

Tightening torque
7—13 N·m {72—132 kgf·cm, 62—115 in·lbf}



am2zzw00010193

Seal rubber installation note
1. Install the seal rubber.
2. Return the wiring harness, air hose component and exhaust gas pressure sensor No.2 that were set aside to their original positions.
3. Connect or install the following parts in the order of the numbers shown in the figure.



am2zzw00010194

(1) Wiring harness protector installation bolts

Tightening torque
8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

(2) Exhaust gas pressure sensor No.2 bracket installation bolts

Tightening torque
8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}

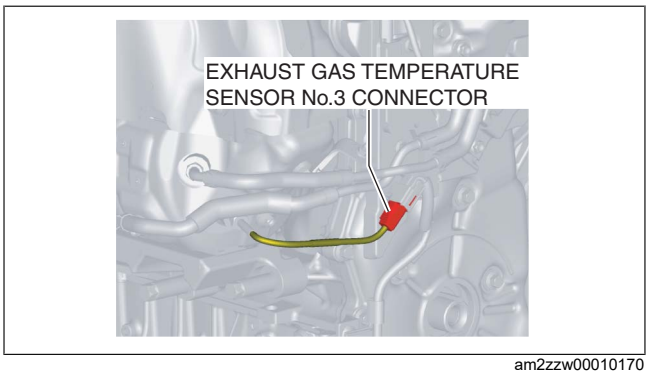
(3) Air pipe stay installation nut

Tightening torque
8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}

(4) Wiring harness clip
(5) Crankshaft position sensor connector
(6) Air hoses

Note
- The seal rubber can be fixed by installing the wiring harness protector.

4. Connect the exhaust gas temperature sensor No.3 connector.

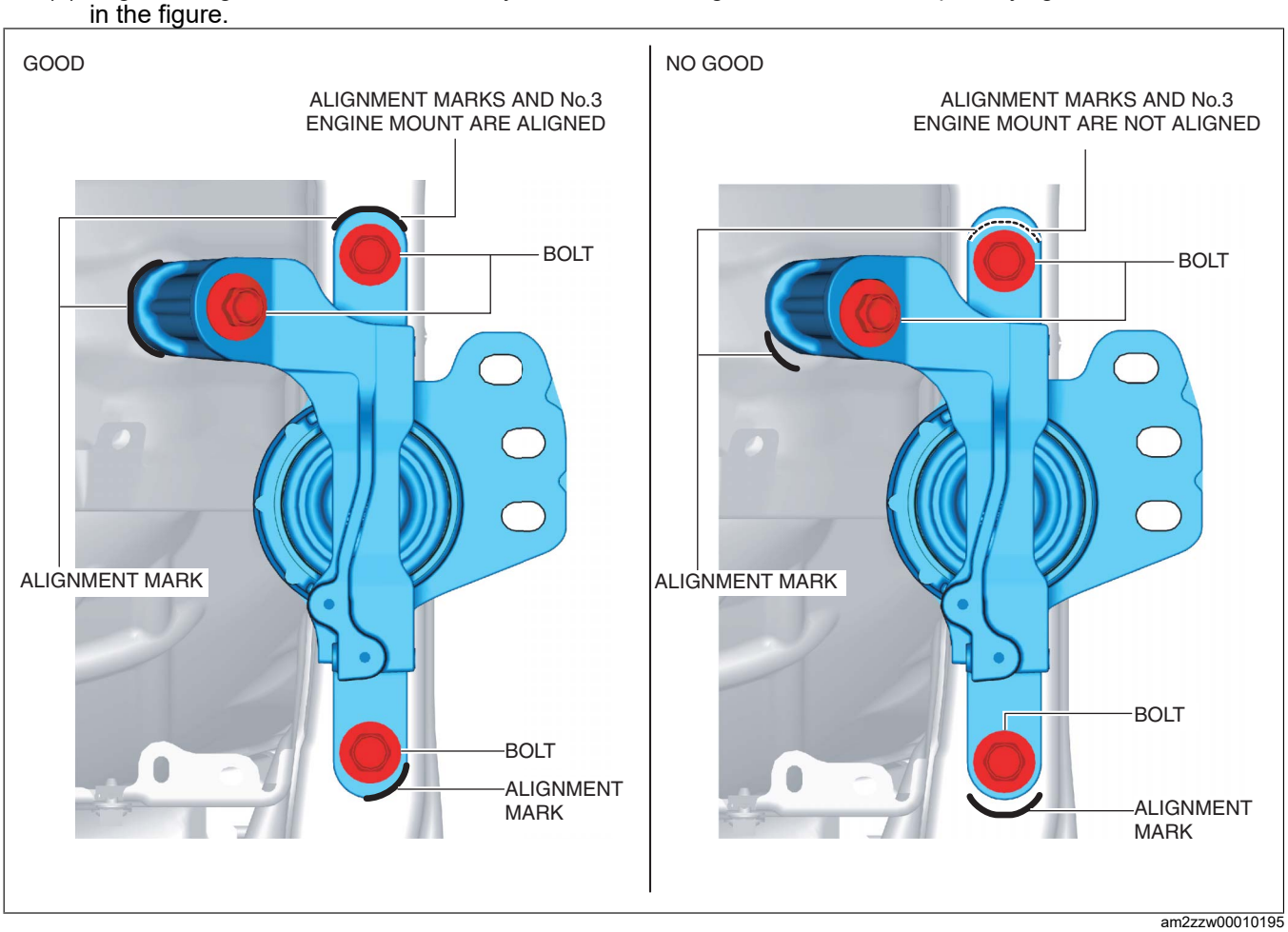


am2zzw00010170

No.3 engine mount installation note

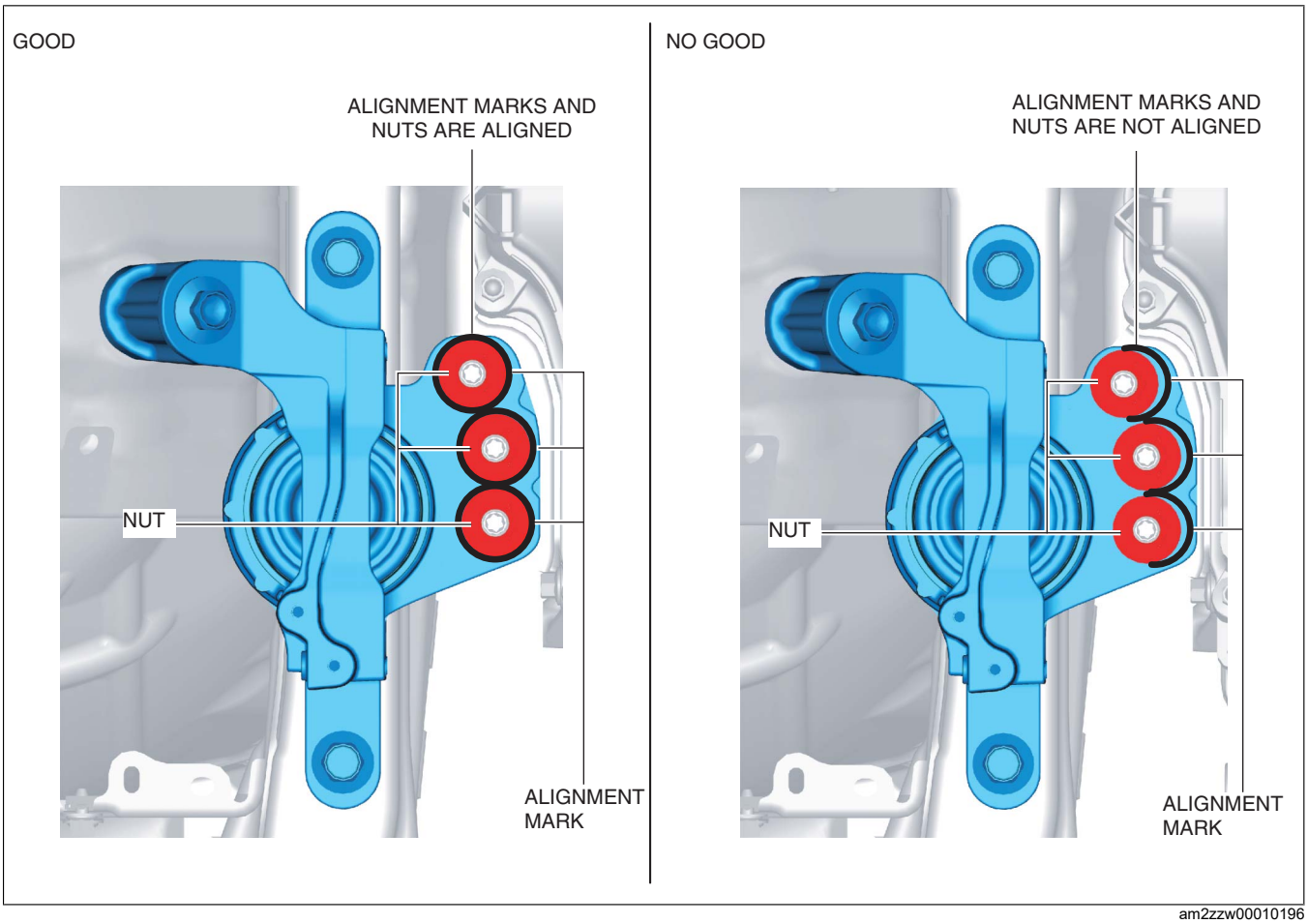
Caution
- **If the No.3 engine mount is deviated from the original position when installing the No.3 engine mount, engine noise or vibration could increase. When installing the No.3 engine mount, align the alignment mark placed during removal and install it to the original position.**

1. Temporarily tighten the No.3 engine mount installation bolts and nuts using the following procedure:
   (1) Align the alignment marks on the body side and No.3 engine mount, and temporarily tighten the bolts shown in the figure.



am2zzw00010195

   (2) Temporarily tighten the nuts shown in the figure while aligning the alignment marks of the No.3 engine mount and nuts.



am2zzw00010196

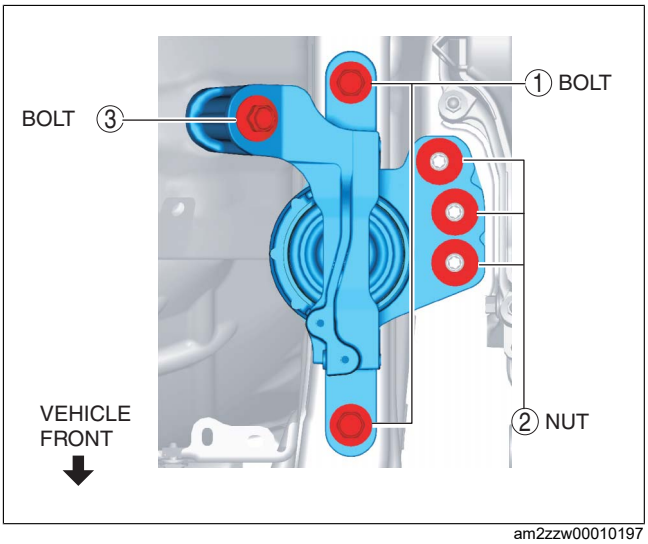
Note

- If the alignment marks are deviated, align the alignment marks while slightly moving the engine and temporarily tighten the nuts.

2. Tighten the No.3 engine mount installation bolts and nuts in the order as shown in the figure.

Tightening torque

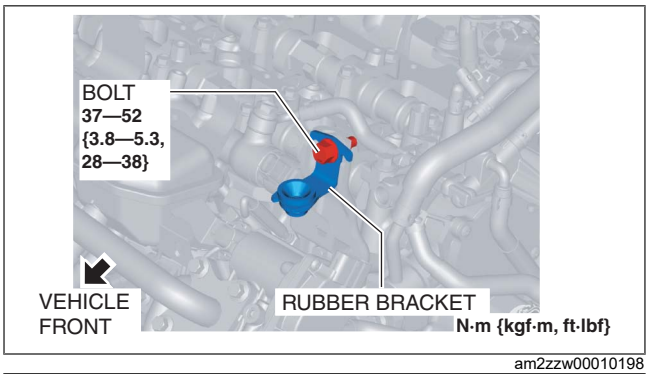
Installation position	Tightening torque
1	76—95 N·m {7.8—9.6 kgf·m, 57—70 ft·lbf}
2	82—95 N·m {8.4—9.6 kgf·m, 61—70 ft·lbf}
3	49—65 N·m {5.0—6.6 kgf·m, 37—47 ft·lbf}



am2zzw00010197

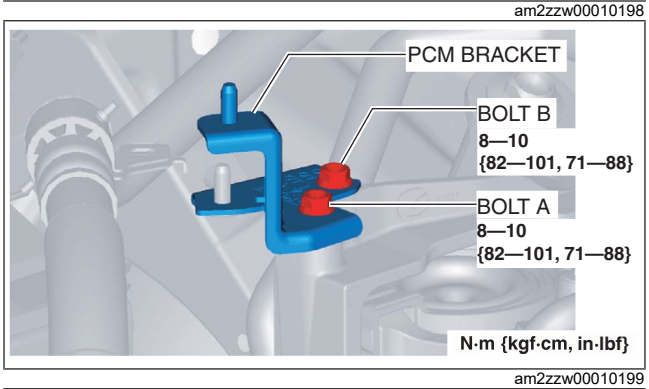
3. Remove the **SSTs**.

4. Install the rubber bracket.



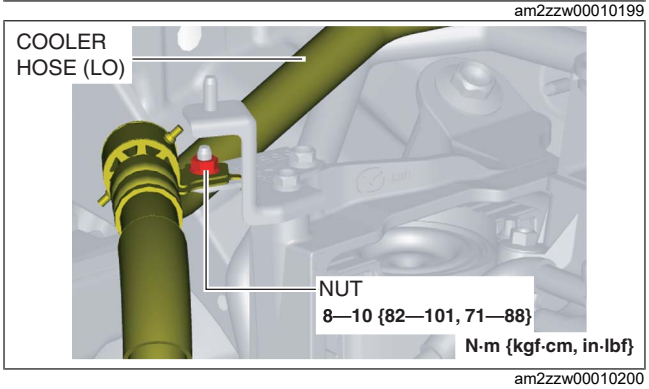
am2zzw00010198

5. Install the PCM bracket using the following procedure:
   (1) Temporarily tighten the bolt A.
   (2) Tighten the bolts in the order of B and A.



am2zzw00010199

6. Install the nut shown in the figure, and then install the cooler hose (LO).



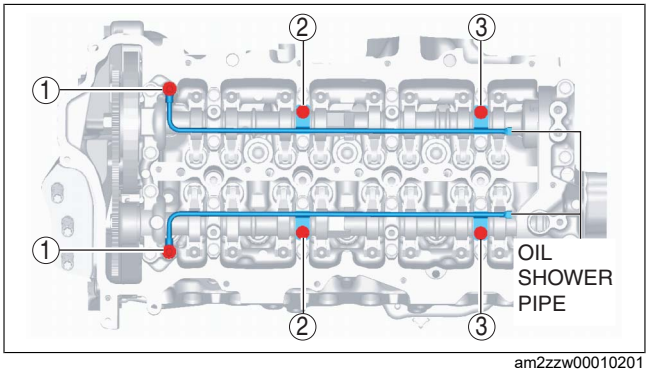
am2zzw00010200

Oil shower pipe installation note

1. Install the oil shower pipe in the order shown in the figure.

Tightening torque

Installation position	Tightening torque
1	9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}
2, 3	8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf}



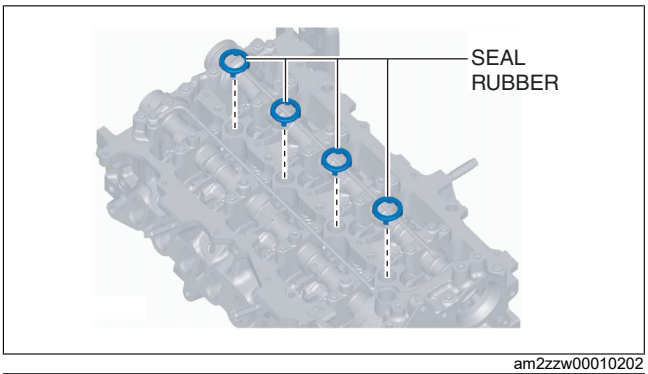
am2zzw00010201

Cylinder head cover installation note

Caution

- **To assure the sealing performance of the cylinder head cover, be careful of the following:**
  - **— Verify that the cylinder head cover gasket is inserted into the cylinder head cover groove and install the cylinder head cover.**
  - **— Completely clean and remove any oil, dirt, sealant or other foreign matter from the seal surface.**

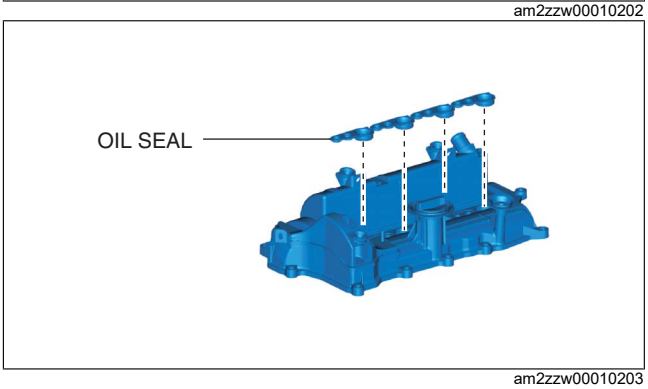
1. Install the seal rubber on the cylinder head.
   - Verify if the seal rubber has been damaged and if so replace it.
     — The seal rubber can be reused even though the cylinder head cover causes contact marks on the seal rubber. However, if there is damage such as cracks and chips which may cause the seal rubber to tear, replace it.



am2zzw00010202

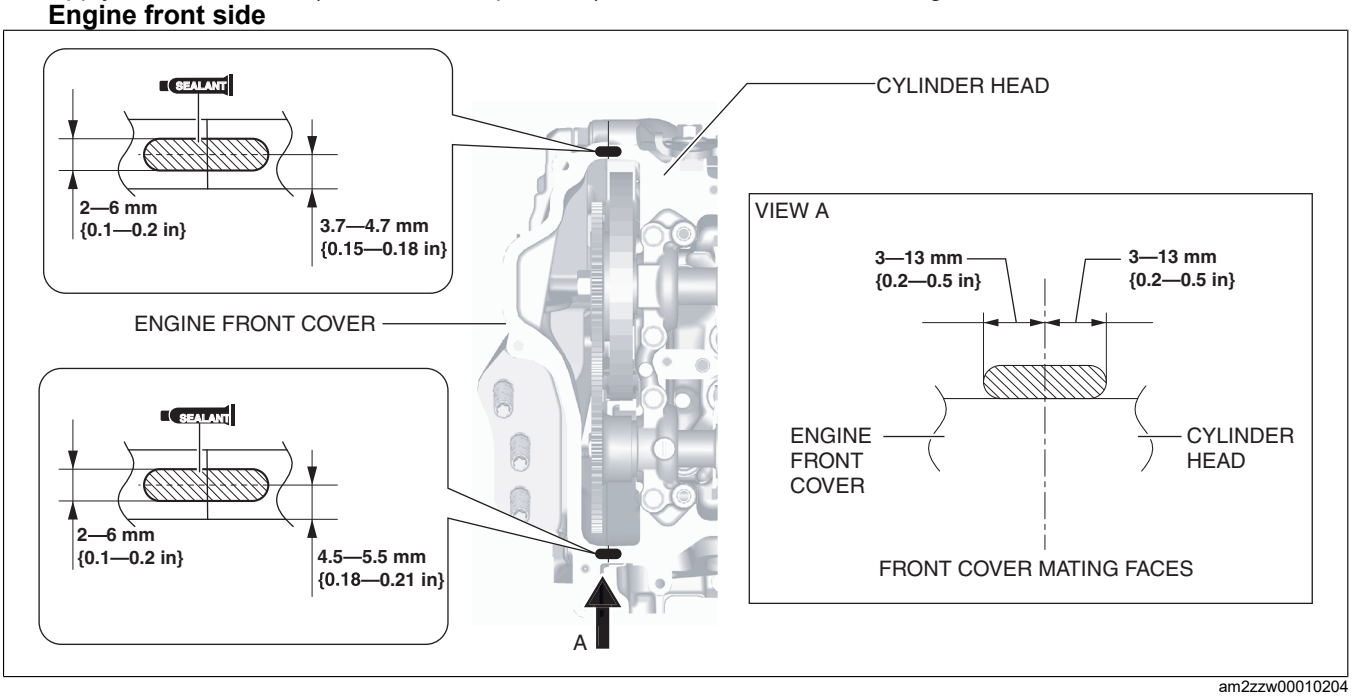
2. Install the oil seal on the cylinder head cover.
   - Verify if the oil seal shown in the figure has been damaged and if so replace it.
3. Insert a new cylinder head cover gasket into the cylinder head cover groove.

Caution
- **To prevent silicone sealant from hardening, adhere the cylinder head cover and the cylinder head firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**



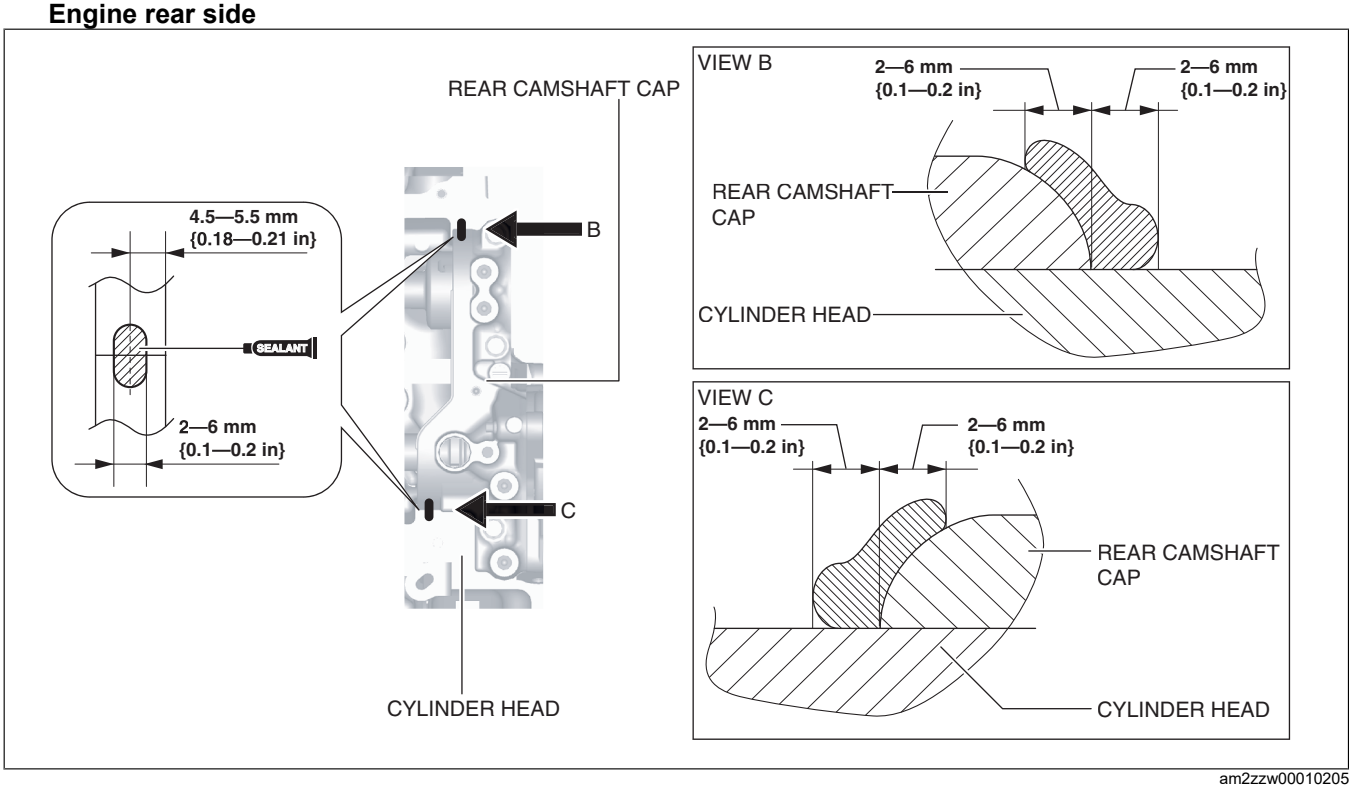
am2zzw00010203

4. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.
   Engine front side



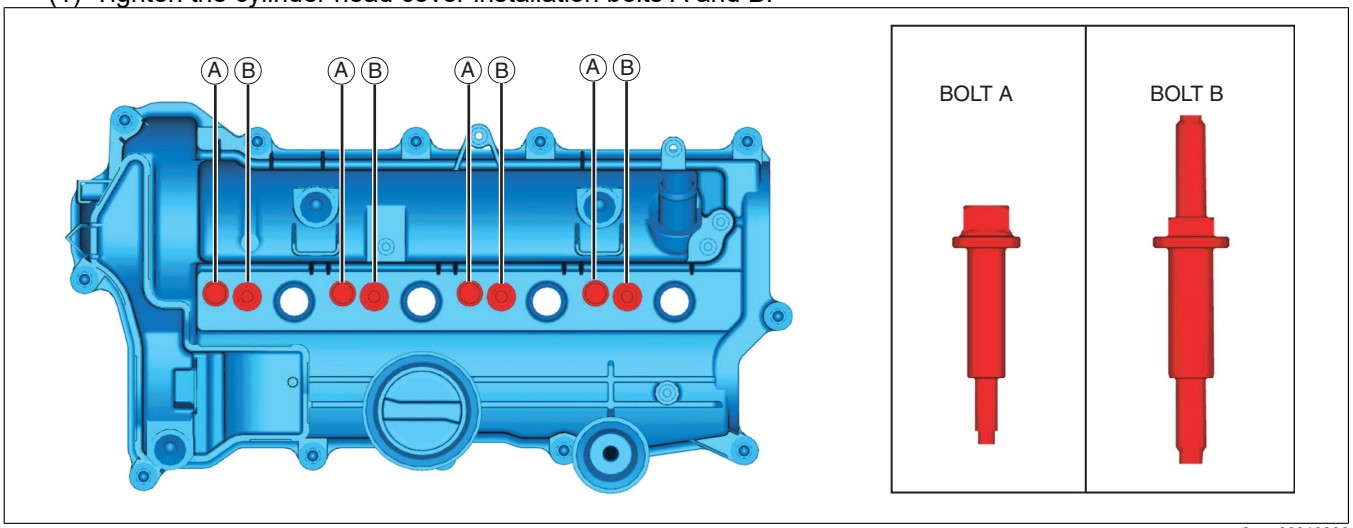
am2zzw00010204

Engine rear side



am2zzw00010205

5. Install the cylinder head cover.
6. Tighten the cylinder head cover installation bolts using the following procedure:
   (1) Tighten the cylinder head cover installation bolts A and B.



am2zzw00010206

Tightening torque

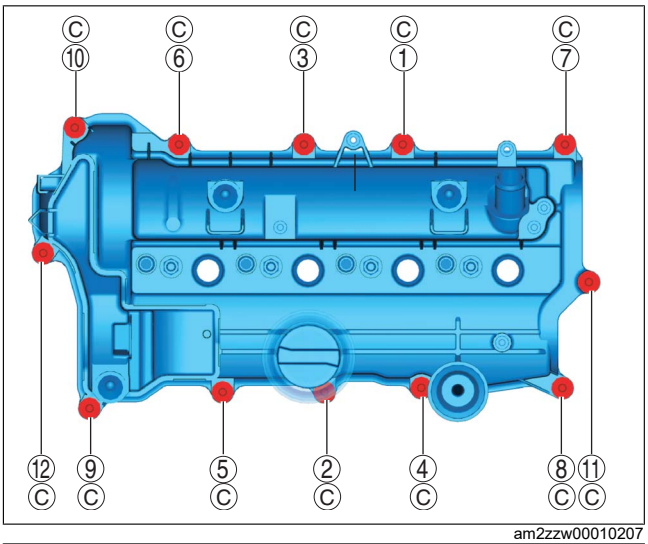
Bolt A: 6.0—8.0 N·m {62—81 kgf·cm, 54—70 in·lbf}
Bolt B: 15—20 N·m {1.6—2.0 kgf·m, 12—14 ft·lbf}

Note

- The tightening order for installation bolts A and B is optional.

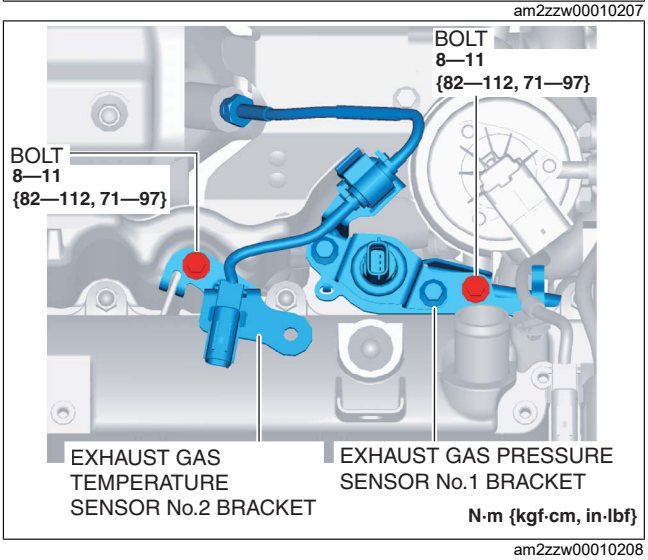
(2) Tighten cylinder head cover installation bolts C in the order shown in the figure.

Tightening torque
4.5—7.0 N·m {46—71 kgf·cm, 40—61 in·lbf}



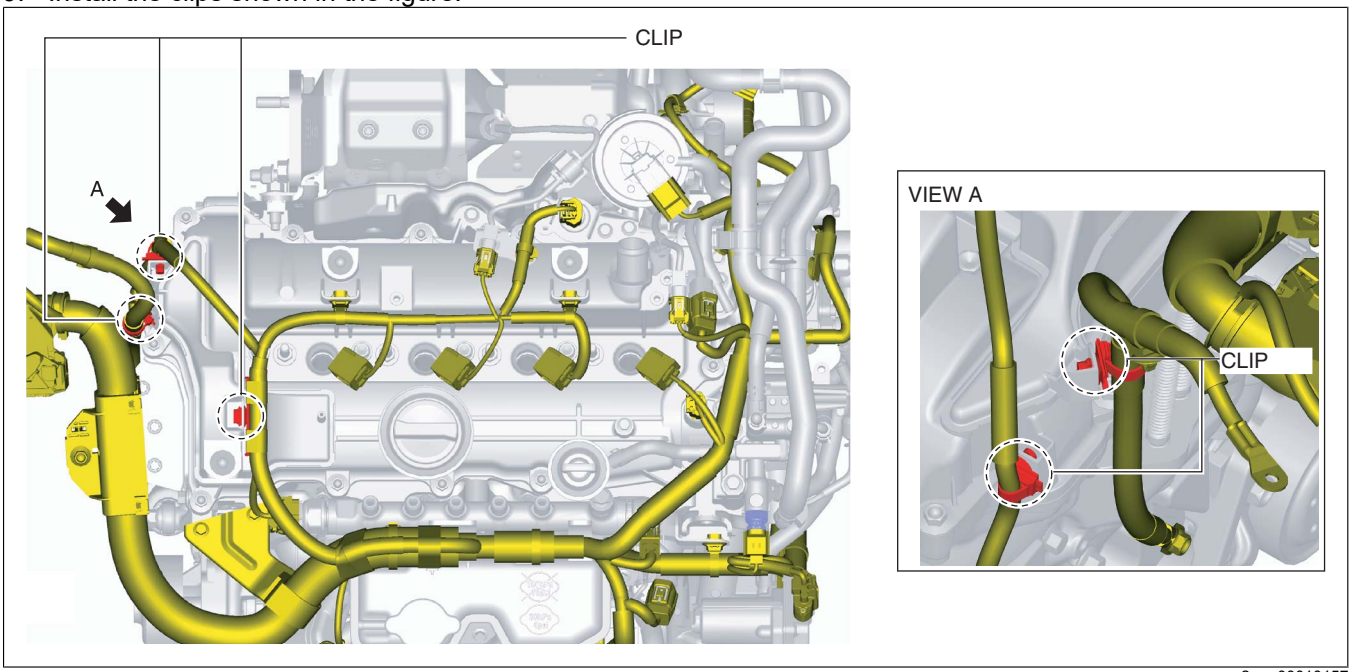
am2zzw00010207

7. Return the exhaust gas pressure No.1 bracket and exhaust gas pressure sensor No.2 bracket to their original positions and tighten the installation bolts.
8. Connect the fuel return hose on the common rail. (See COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



am2zzw00010208

9. Install the clips shown in the figure.



am2zzw00010157

10. Install the vacuum pipe. (See SUPPLY PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
11. Connect the breather hose to the cylinder head cover. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

12. Install the ignition pipe (supply pump side). (See INJECTION PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)